

BridgeU
Software Design Development

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Chapter 1 Introduction

1.1 Purpose and Scope

The purpose of the software design development is to present a complete software design of the BridgeU project. This document contains the outline of the software architecture, functions used, and activity diagrams with the user interface for this project. BridgeU aims to eliminate language barriers through AI technology and establish a bilingual mutual assistance ecosystem connecting Chinese students, global students, and local Thai merchants.

1.2 Acronyms and Definitions

Acronyms

SRS	Software Requirement Specification
UI	User Interface
SDD	Software Design Document
FD	Function Description

Name	Definition
IEEE	Institute for Electrical and Electronics Engineers. Biggest global interest group for engineers of different branches and computer scientists.
Plan	A document series of tasks requires meeting an objective, typically including the associated schedule, budget, resources, organizational description and work breakdown structure.
Feature	Feature Transformation of input parameters to output parameters based

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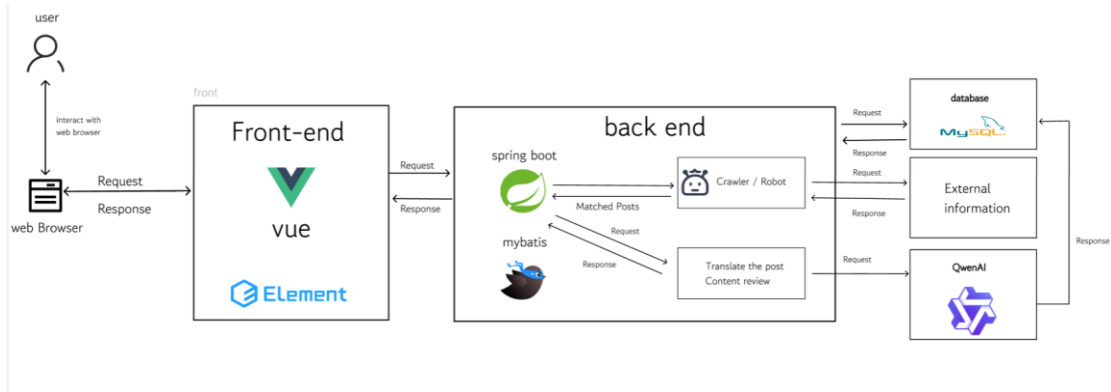
	on a specified algorithm. It describes the functionality of the product in the language of the product. Used for requirements analysis, design, coding, testing or maintenance.
Project Management	The application of knowledge, skills, tools, and techniques to project activities in order to meet or exceed stakeholder needs and expectations from a project.
Project Plan	A formal, approved document used to guide both project execution and project control. The primary uses of the project plan are to document planning assumptions and the decision, to facilitate communication among stakeholders, and to document approved scope, cost, and schedule baseline.
Risk	An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives. It is a function of the probability of occurrence of a given threat's occurrence.
Risk Management	The systematic application of management policies, procedures and practices to the tasks of identifying, analyzing, evaluating, treating and monitoring risk.
System Testing	Testing conducted on a complete and integrated system for evaluate the system's compliance with its specified requirements.
Unit Test	A test of individual programs or modules in order to remove a design or programming errors.

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Traceability	The ability to trace the history, application or location of an item or activity, or work products or activities, by means of recorded identification. The establishment and maintenance of relationships between such items. Horizontal traceability describes the relationship between work products of the same type (e.g. Customer requirements). Vertical traceability describes the relationship between work products, which build or derived from each other (e.g., from customer requirements to qualification test cases). Bidirectional traceability allows to directly following relationships in both directions.
Use Case	A use case is a description of a specific interaction between a system and its users to accomplish a particular goal or task. It outlines the sequence of steps or actions that a user performs in the system and describes the system's responses to those actions. Use cases help to identify and define the functional requirements of a system from the perspective of its users.
Crawler	A program that systematically browses the World Wide Web in order to create an index of data.
Qwen AI	he Large Language Model provided by Alibaba Cloud used for translation and summarization.

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Chapter 2 System Architecture



Description:

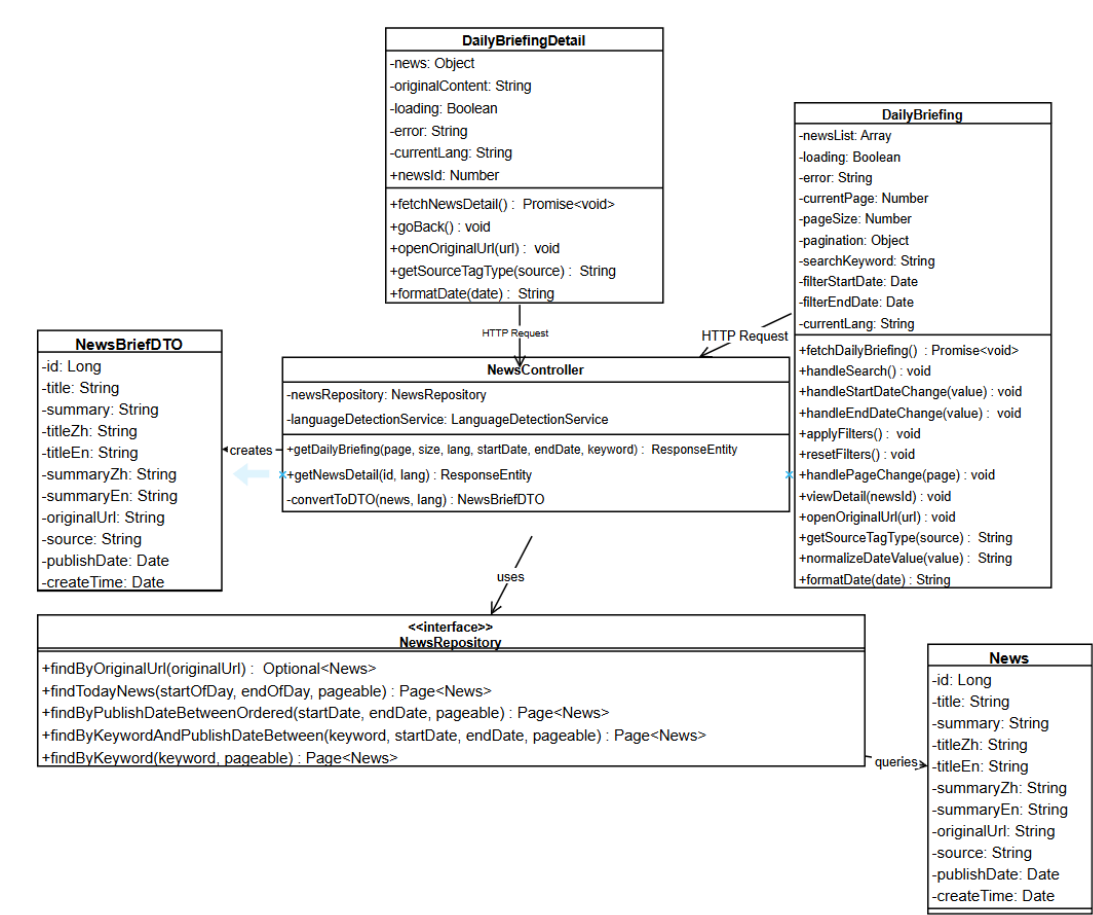
The system architecture consists of a front-end built using Vue.js 3 and ElementPlus 2.6, which provides a responsive user interface. The back-end is built with Spring Boot3.2, a robust Java framework, handling business logic and RESTful API requests. MySQL 8 is used as the primary database for storing structured user and post data. Elasticsearch is integrated to provide high-performance semantic search capabilities. Additionally, the system connects to the Aliyun Qwen3 max API to perform natural language processing tasks such as summarization and translation. Axios is used for communication between the front-end and back-end.

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Chapter 3 Detail Design

3.1 Feature 1

3.1.1 Class Diagram



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3.1.2 Class Diagram Description

3.1.2.1 View

3.1.2.1.1 DailyBriefing

DailyBriefing
-newsList: Array -loading: Boolean -error: String -currentPage: Number -pageSize: Number -pagination: Object -searchKeyword: String -filterStartDate: Date -filterEndDate: Date -currentLang: String
+fetchDailyBriefing() : Promise<void> +handleSearch() : void +handleStartDateChange(value) : void +handleEndDateChange(value) : void +applyFilters() : void +resetFilters() : void +handlePageChange(page) : void +viewDetail(newsId) : void +openOriginalUrl(url) : void +getSourceTagType(source) : String +normalizeDateValue(value) : String +formatDate(date) : String

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Attributes:

ID	Name	Description	Type
1	newsList	Array containing the list of Daily Briefing news items to be displayed	Array
2	loading	Boolean flag indicating if the news data is currently being fetched	Boolean
3	error	String containing error message if data fetching fails	String
4	currentPage	Current page number for pagination (1-based)	Number
6	pageSize	Number of news items to display per page	Number
7	searchKeyword	String containing the keyword entered by the user for searching news	String
8	filterStartDate	Date object representing the start date for filtering news by publication date	Date
9	filterEndDate	Date object representing the end date for filtering news by publication date	Date
10	currentLang	String representing the current interface language ("zh" for Chinese, "en" for English)	String

Methods:

ID	Name	Description	Parameter	Return Type
1	fetchDailyBriefing	Fetches Daily Briefing news items	None	Promise<void>

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		from the backend API with pagination, search, and filter parameters. Sets loading=true, calculates 0-based page number, builds params object with page, size, lang, keyword (if provided), startDate and endDate. Sends GET request to /api/news/daily-briefing. On success, updates newsList and pagination object. Handles errors and sets loading=false in finally block.		
2	handleSearch	Handles search action by resetting currentPage to 1 and calling fetchDailyBriefing(). This ensures search results start from the first page.	None	Void
3	handleStartDateChange	Updates the filterStartDate when user selects a start date	value: Date \null	void

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4	handleEndDateChange	Updates the filterEndDate when user selects an end date	value: Date \null	void
5	applyFilters	Validates date range (checks if startDate > endDate, shows warning if invalid). If valid, resets currentPage to 1 and calls fetchDailyBriefing() to fetch filtered results. Note: fetchDailyBriefing() internally calls normalizeDateValue() to normalize date values before sending to API.	None	void
6	resetFilters	Clears all filter conditions: sets searchKeyword="", filterStartDate=null, filterEndDate=null, resets currentPage to 1, and calls fetchDailyBriefing() to reload all news without filters.	None	void
7	handlePageChange	Handles pagination by updating currentPage to the new page number and	page: Number	void

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		calling fetchDailyBriefing() to fetch the corresponding page. Scrolls to top after page change.		
8	viewDetail	Emits an event to navigate to the news detail page for a specific news item	newsId: Number	void
9	openOriginalUrl	Opens the original news article URL in a new browser tab	url: String	void
10	getSourceTagType	Returns the Element UI tag type (primary/success/info) based on the news source	source: String	String
11	normalizeDateValue	Normalizes date value to DD-MM-YYYY format string. Handles Date objects (extracts year, month, day), string values (validates DD-MM- YYYY format, ignores incomplete inputs with letters), and null values. Returns null for invalid inputs.	value: Date \String	String \null

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	formatDate	Formats a date object to "DD-MM-YYYY HH:mm" format string	date: Date \String	String
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3.1.2.1.2 DailyBriefingDetail

DailyBriefingDetail
-news: Object -originalContent: String -loading: Boolean -error: String -currentLang: String +newsId: Number
+fetchNewsDetail() : Promise<void> +goBack() : void +openOriginalUrl(url) : void +getSourceTagType(source) : String +formatDate(date) : String

Attributes:

ID	Name	Description	Type
1	news	Object containing the detailed information of a single Daily Briefing news item	Object

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2	originalContent	String containing the original content of the news article	String
3	loading	Boolean flag indicating if the news detail data is currently being fetched	Boolean
4	error	String containing error message if data fetching fails	String
5	currentLang	String representing the current interface language ("zh" for Chinese, "en" for English)	String
6	newsId	Number representing the ID of the news item to display (passed as prop)	Number

Methods:

ID	Name	Description	Parameter	Return Type
1	fetchNewsDetail	Fetches detailed information of a single news item from the backend API using newsId. Sets loading=true, error=null, builds params with lang (currentLang or 'en'), sends GET request to /api/news/daily-briefing/{id}. First checks response.status === 200, then checks response.data.success. On success, updates news = response.data.data and originalContent = response.data.originalContent	None	Promise<void>

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		t. Handles errors: 404 (shows localized notFound message), 401 (shows login expired message, handled by interceptor), 500 (shows error message from response), network errors (shows localized networkError message). Sets loading=false in finally block.		
2	goBack	Emits an event to navigate back to the Daily Briefing list page	None	Void
3	openOriginalUrl	Opens the original news article URL in a new browser tab	url: String	Void
4	getSourceTagType	Returns the Element UI tag type (primary/success/info) based on the news source	source: String	String
5	formatDate	Formats a date object to "DD-MM-YYYY HH:mm" format string	date: Date \String	String

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3.1.2.2 Controller

3.1.2.2.1 Controller: NewsController

NewsController	
*-newsRepository: NewsRepository	
-languageDetectionService: LanguageDetectionService	
+getDailyBriefing(page, size, lang, startDate, endDate, keyword) : ResponseEntity	
+getNewsDetail(id, lang) : ResponseEntity	
-convertToDTO(news, lang) : NewsBriefDTO	

Attributes:

ID	Name	Description	Type
1	newsRepository	Instance of NewsRepository for database operations	NewsRepository
2	languageDetectionService	Instance of LanguageDetectionService for detecting Thai and Chinese content	LanguageDetectionService

Methods:

ID	Name	Description	Parameter	Return Type
1	getDailyBriefing	Handles GET request to retrieve paginated Daily Briefing news list with optional filters (keyword, date range, source).	page: int, size: int, lang: String, source: String,	ResponseEntity<Map<String, Object>>

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		Returns ResponseEntity with paginated news data.	startDate: String, endDate: String, keyword: String	
2	convertToDTO	Converts News entity to NewsBriefDTO based on language preference. Ensures no Thai content is displayed on the website.	news: News, lang: String	NewsBriefDTO
3	getNewsDetail	Handles GET request to retrieve detailed information of a single news item by ID. Returns DTO data and optional original content.	`id: Long`, `lang: String`	`ResponseEntity<Map<String, Object>>`

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3.1.2.3 Repository

3.1.2.3.1 Repository: NewsRepository

<<interface>> NewsRepository	
+findByOriginalUrl(originalUrl) : Optional<News>	
+findTodayNews(startOfDay, endOfDay, pageable) : Page<News>	
+findByPublishDateBetweenOrdered(startDate, endDate, pageable) : Page<News>	
+findByKeywordAndPublishDateBetween(keyword, startDate, endDate, pageable) : Page<News>	
+findByKeyword(keyword, pageable) : Page<News>	

Methods:

ID	Name	Description	Parameter	Return Type
1	findByOriginalUrl	Finds a news item by its original URL (for deduplication)	originalUrl: String	Optional<News>
2	findTodayNews	Finds today's news items based on publishDate (with pagination)	startOfDay: Date, endOfDay: Date, pageable: Pageable	Page<News>
3	findByPublishDateBetweenOrdered	Finds news items within publish date range, ordered by publishDate DESC (with pagination)	startDate: Date, endDate: Date, pageable: Pageable	Page<News>
4	findByKeywordAndPublishDateBetween	Searches news items by keyword (in title, content, summary, translations) and publish date range (with pagination)	keyword: String, startDate: Date, endDate: Date,	Page<News>

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			pageable: Pageable	
5	findByKeyword	Searches news items by keyword across all historical news (with pagination)	keyword: String, pageable: Pageable	Page<News>

3.1.2.4 Model

3.1.2.4.1 Model: News

News
-id: Long -title: String -summary: String -titleZh: String -titleEn: String -summaryZh: String -summaryEn: String -originalUrl: String -source: String -publishDate: Date -createTime: Date

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Attributes:

ID	Name	Description	Type
1	Id	Primary key ID of the news item	Long
2	title	Original news title	String
3	createTime	timestamp when the news was crawled and stored	Date
4	summary	AI-generated summary of the news article	String
5	titleZh	Chinese translation of the title	String
6	titleEn	English translation of the title	String
7	summaryZh	Chinese translation of the summary	String
8	summaryEn	English translation of the summary	String
9	originalUrl	Original article URL from the source website	String
10	source	Source website name	String
11	publishDate	Publication date from the source website	Date

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3.1.2.4.2 Model: NewsBriefDTO

NewsBriefDTO
-id: Long
-title: String
-summary: String
-titleZh: String
-titleEn: String
-summaryZh: String
-summaryEn: String
-originalUrl: String
-source: String
-publishDate: Date
-createTime: Date

Attributes:

ID	Name	Description	Type
1	Id	Primary key ID of the news item	Long
2	Title	news title (language-specific based on user preference)	String
3	Summary	news summary (language-specific based on user preference)	String
4	titleZh	Chinese translation of the title	String
5	titleEn	English translation of the title	String
6	summaryZh	Chinese translation of the summary	String

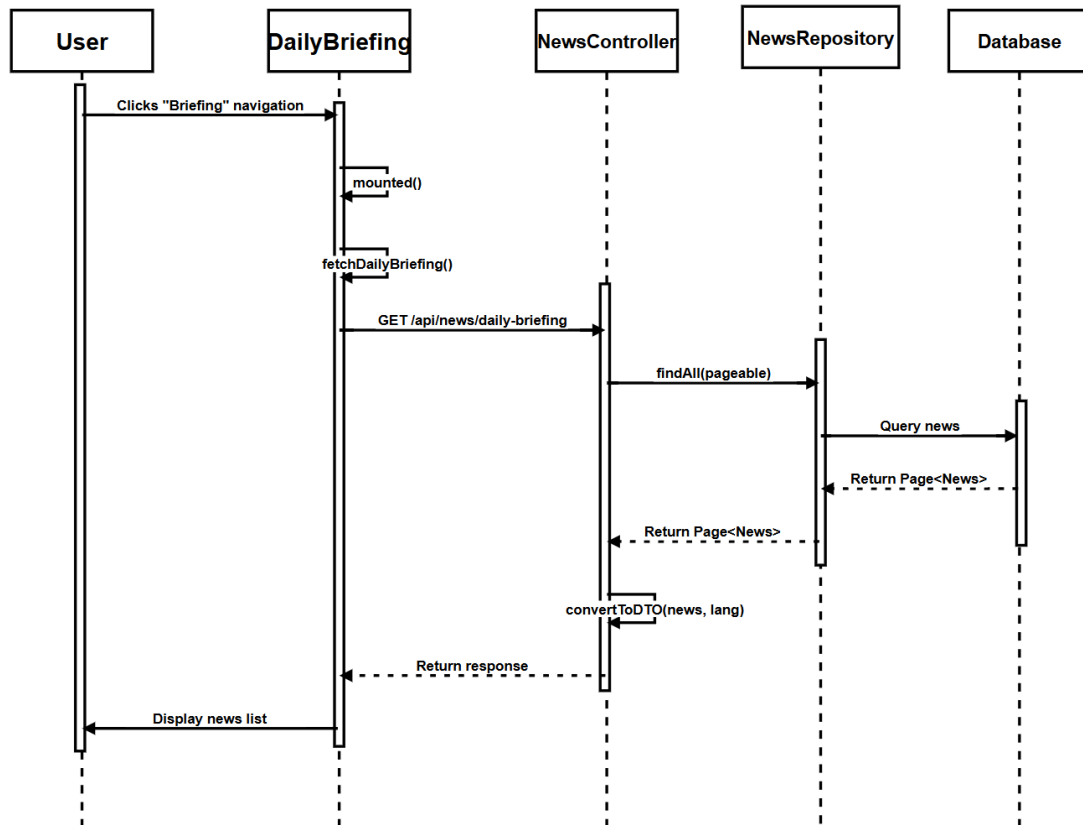
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7	summaryEn	English translation of the summary	String
8	originalUrl	Original article URL from the source website	String
9	source	Source website name	String
10	publishDate	Publication date from the source website	Date
11	createTime	Timestamp when the news was crawled and stored	Date

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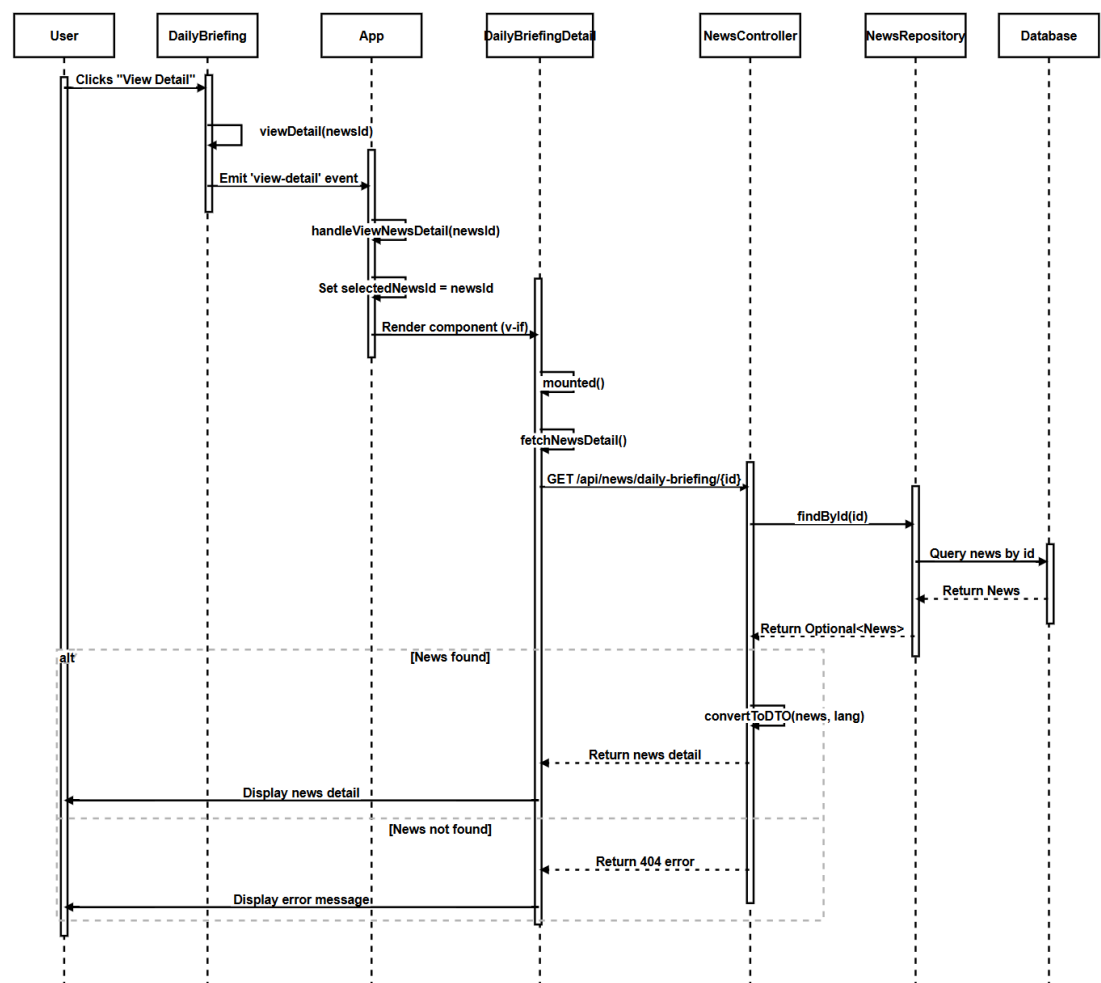
3.1.3 Sequence Diagram

SD-01: View Daily Briefings



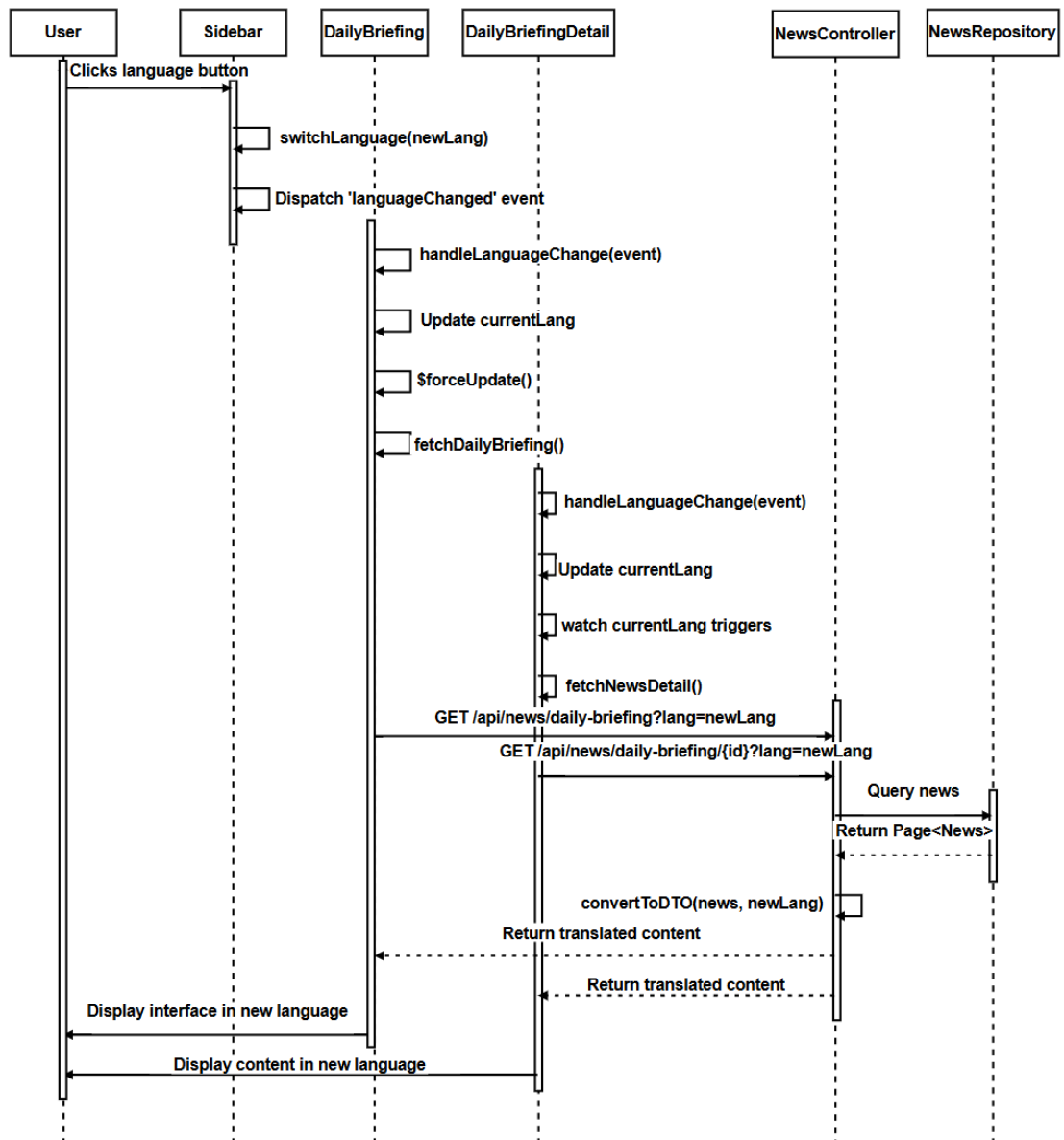
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SD-02: View Daily Briefing News Details



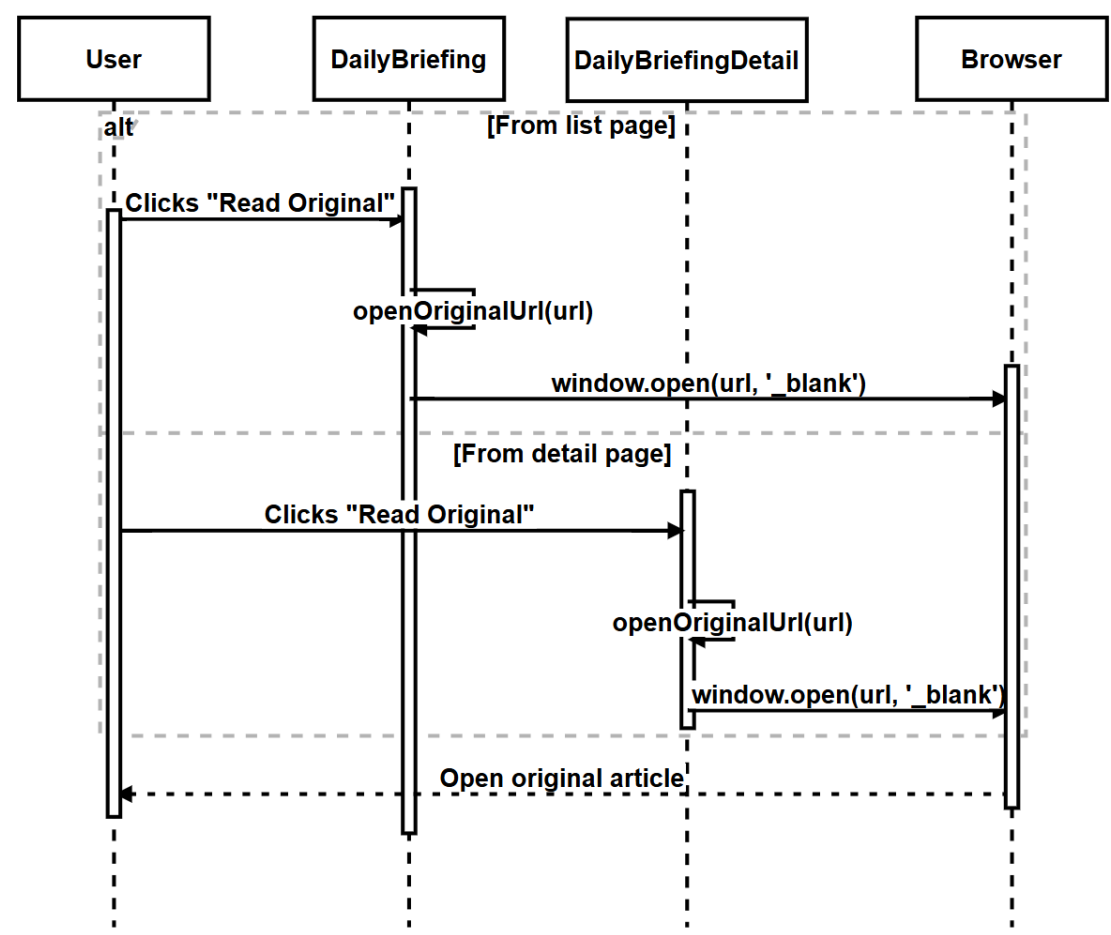
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SD-03: Switch Interface Language



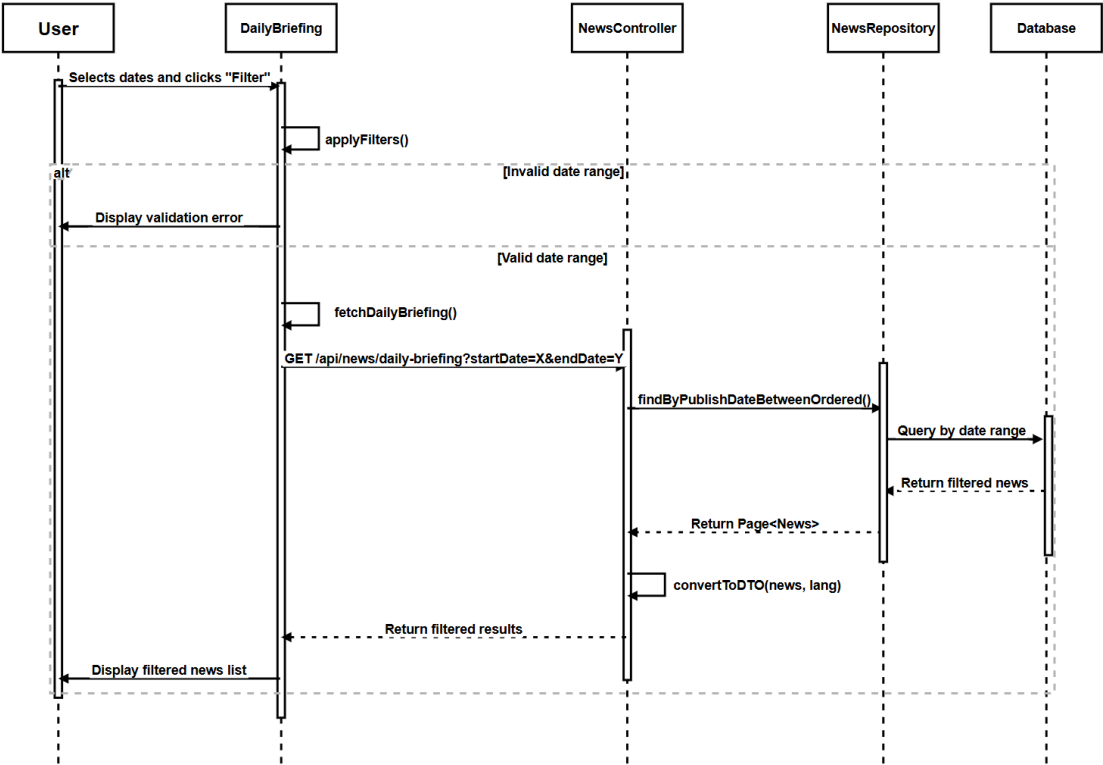
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SD-04: Jump to Original Link



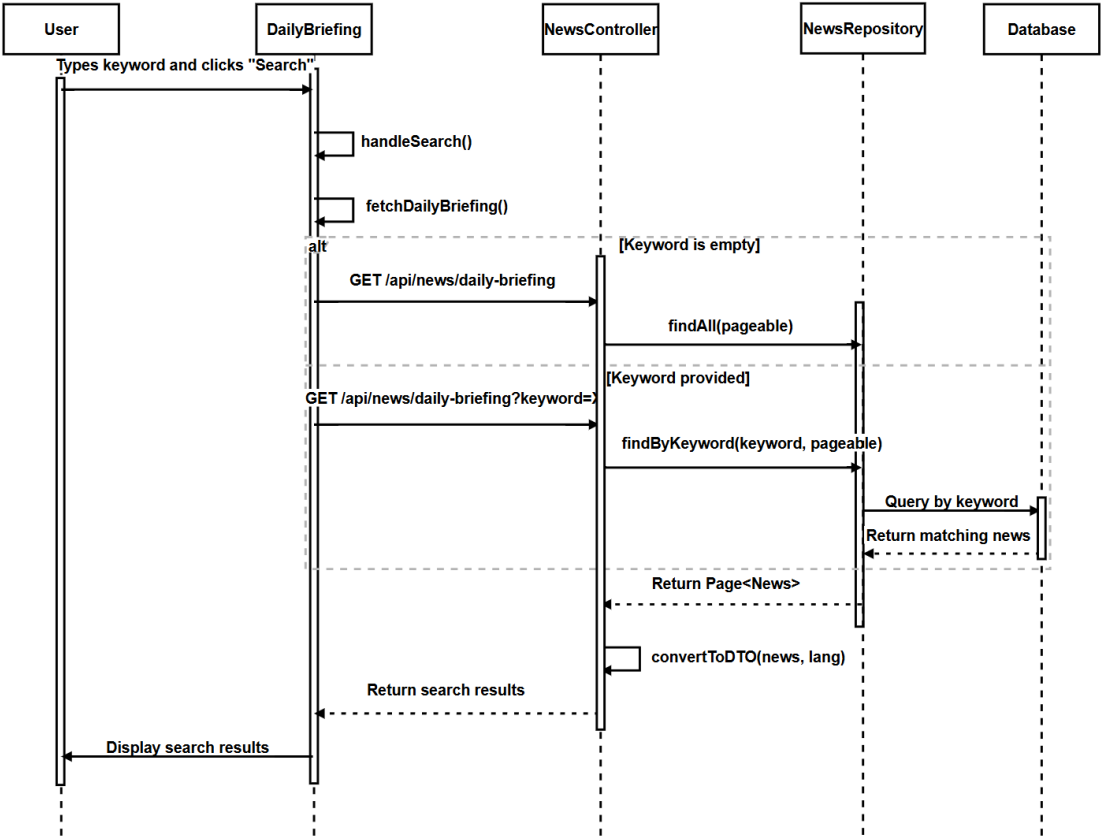
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SD-05: Filter Daily Briefing News by Date



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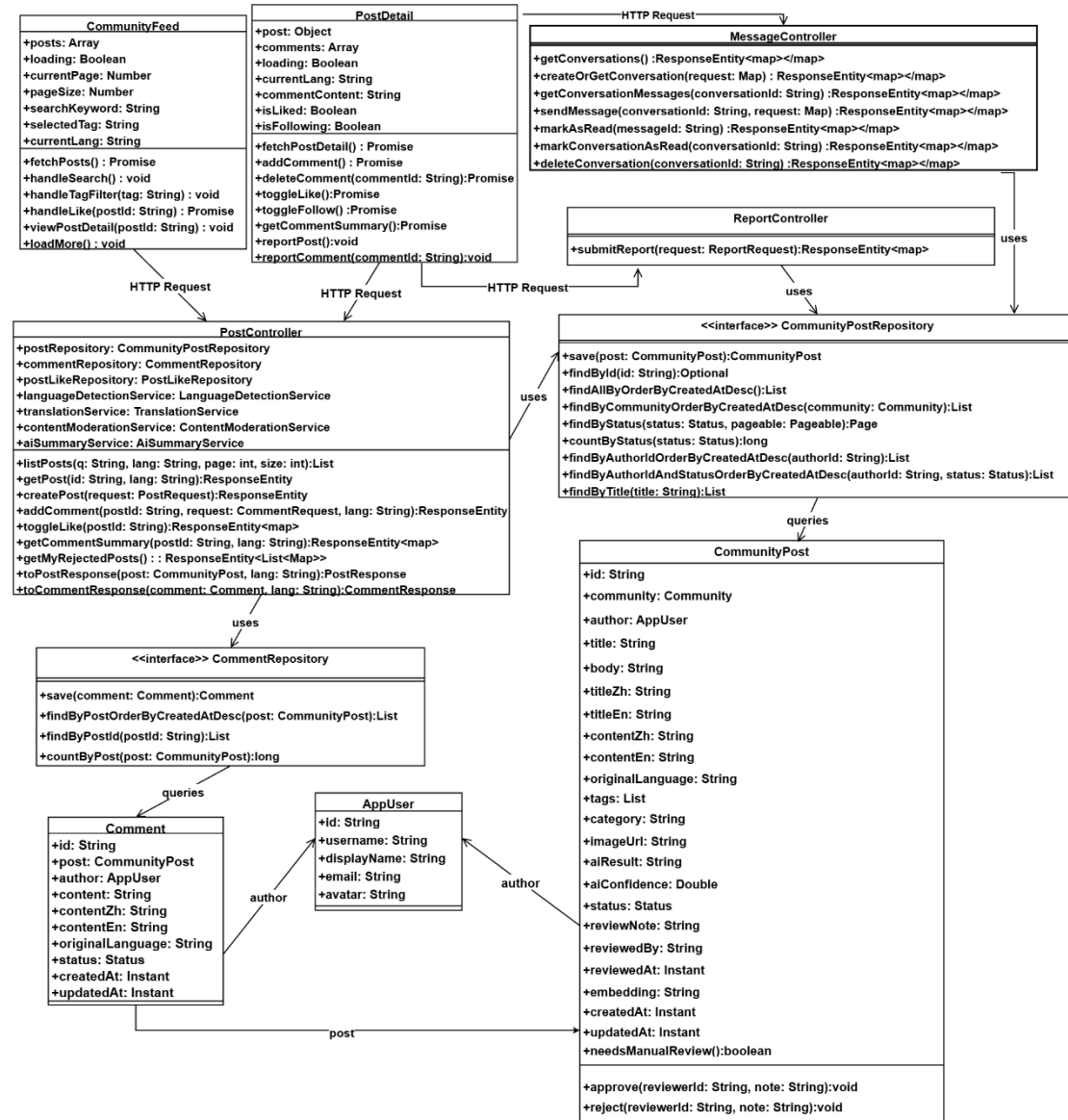
SD-06: Search Daily Briefing News



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3.2 Feature 2

3.2.1 Class Diagram



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3.2.2 Class Diagram Description

3.2.2.1 View

3.2.2.1.1 CommunityFeed

CommunityFeed
+posts: Array +loading: Boolean +currentPage: Number +pageSize: Number +searchKeyword: String +selectedTag: String +currentLang: String
+fetchPosts() : Promise +handleSearch() : void +handleTagFilter(tag: String) : void +handleLike(postId: String) : Promise +viewPostDetail(postId: String) : void +loadMore() : void

Attributes:

ID	Name	Description	Type
1	posts	Community posts currently displayed in the feed	Array
2	loading	Whether posts are currently being fetched	Boolean
3	currentPage	Current page number for pagination	Number

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4	pageSize	Number of posts to display per page	Number
5	searchKeyword	Keyword used to search posts	String
6	selectedTag	Currently selected tag filter (Study, Housing, Travel, Part-time Job, Life Services)	String
7	currentLang	Current interface language ("zh" for Chinese, "en" for English)	String

Methods:

ID	Name	Description	Parameter	Return Type
1	fetchPosts	Fetches community posts from the backend API with current filters and language	None	Promise<void>
2	handleSearch	Updates `searchKeyword` and reloads posts	None	void
3	handleTagFilter	Updates `selectedTag` and reloads posts	`tag: String`	void
4	handleLike	Sends like/unlike request for a post	`postId: String`	Promise<void>
5	viewPostDetail	Navigates to the post detail page for a specific post	`postId: String`	void
6	loadMore	Loads the next page of posts and appends them to `posts`	None	void

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3.2.2.1.2 PostDetail

PostDetail
+post: Object +comments: Array +loading: Boolean +currentLang: String +commentContent: String +isLiked: Boolean +isFollowing: Boolean
+fetchPostDetail() : Promise +addComment() : Promise +deleteComment(commentId: String):Promise +toggleLike():Promise +toggleFollow() :Promise +getCommentSummary():Promise +reportPost():void +reportComment(commentId: String):void

Attributes:

ID	Name	Description	Type
1	post	Detailed information of the selected post	Object
2	comments	Comments for the post	Array
3	loading	Whether post detail data is currently being fetched	Boolean
4	currentLang	Current interface language ("zh" for Chinese, "en" for English)	String
5	commentContent	Content of a new comment being composed	String
6	isLiked	Whether the current user has liked this post	Boolean

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7	isFollowing	Whether the current user is following the post author	Boolean
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Methods:

ID	Name	Description	Parameter	Return Type
1	fetchPostDetail	Fetches detailed information of the post and its comments	None	Promise<void>
2	addComment	Sends a request to add a new comment and refreshes the comments list	None	Promise<void>
3	deleteComment	Sends a request to delete a comment	`commentId: String`	Promise<void>
4	toggleLike	Sends a like/unlike request for the post	None	Promise<void>
5	toggleFollow	Sends a follow/unfollow request for the post author	None	Promise<void>
6	getCommentSummary	Requests an AI-generated summary of all comments for the post	None	Promise<void>
7	reportPost	Sends a report request for the post	None	void
8	reportComment	Sends a report request for a specific comment	`commentId: String`	void

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3.2.2.2 Controller

3.2.2.2.1 Controller: `PostController`

PostController
+postRepository: CommunityPostRepository +commentRepository: CommentRepository +postLikeRepository: PostLikeRepository +languageDetectionService: LanguageDetectionService +translationService: TranslationService +contentModerationService: ContentModerationService +aiSummaryService: AiSummaryService
+listPosts(q: String, lang: String, page: int, size: int):List +getPost(id: String, lang: String):ResponseEntity +createPost(request: PostRequest):ResponseEntity +addComment(postId: String, request: CommentRequest, lang: String):ResponseEntity +toggleLike(postId: String):ResponseEntity<map> +getCommentSummary(postId: String, lang: String):ResponseEntity<map> +getMyRejectedPosts() : : ResponseEntity<List<Map>> +toPostResponse(post: CommunityPost, lang: String):PostResponse +toCommentResponse(comment: Comment, lang: String):CommentResponse

Attributes:

I D	Name	Description	Type
1	postRepository	Instance of 'CommunityPostRepository' for database operations	CommunityPostRepository
2	commentRepository	Instance of 'CommentRepository' for comment operations	CommentRepository
3	postLikeRepository	Instance of 'PostLikeRepository' for like operations	postLikeRepository

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4	languageDetectionService	Instance of 'LanguageDetectionService' for language detection	LanguageDetectionService
5	translationService	Instance of 'TranslationService' for content translation	TranslationService
6	contentModerationService	Instance of 'ContentModerationService' for AI content moderation	ContentModerationService
7	aiSummaryService	Instance of 'AiSummaryService' for AI-generated summaries	AiSummaryService

Methods:

ID	Name	Description	Parameter	Return Type
1	listPosts	Returns a paginated list of community posts with optional filters (keyword, tag, language)	'q: String', 'lang: String', 'page: int', 'size: int'	'List<PostListResponse>'
2	getPost	Returns detailed information of a single post including comments	'id: String', 'lang: String'	'ResponseEntity<PostDetailResponse>'
3	createPost	Creates a new community post with AI moderation and translation	'request: PostRequest'	'ResponseEntity<PostResponse>'

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4	addComment	Adds a new comment to the specified post with translation	`postId: String`, `request: CommentRequest`, `lang: String`	`ResponseEntity<CommentResponse>`
5	toggleLike	Toggles like status for a post	`postId: String`	`ResponseEntity<Map>`
6	getCommentSummary	Returns an AI-generated summary of all comments for a post	`postId: String`, `lang: String`	`ResponseEntity<Map>`
7	getMyRejectedPosts	Returns list of rejected posts for the current user	None	`ResponseEntity<List<Map>>`
8	toPostResponse	Converts `CommunityPost` entity to `PostResponse` DTO based on language preference	`post: CommunityPost`, `lang: String`	`PostResponse`
9	toCommentResponse	Converts `Comment` entity to `CommentResponse` DTO based on language preference	`comment: Comment`, `lang: String`	`CommentResponse`

3.2.2.2.2 Controller: `ReportController`

ReportController
+submitReport(request: ReportRequest):ResponseEntity<map>

Methods:

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ID	Name	Description	Parameter	Return Type
1	submitReport	Submits a new report for a post or comment	`request: ReportRequest`	`ResponseEntity<Map>`

3.2.2.2.3 Controller: `MessageController`

MessageController
+getConversations() :ResponseEntity<map></map> +createOrGetConversation(request: Map) : ResponseEntity<map></map> +getConversationMessages(conversationId: String) :ResponseEntity<map></map> +sendMessage(conversationId: String, request: Map) :ResponseEntity<map></map> +markAsRead(messageId: String) :ResponseEntity<map></map> +markConversationAsRead(conversationId: String) :ResponseEntity<map></map> +deleteConversation(conversationId: String) :ResponseEntity<map></map>

Methods:

ID	Name	Description	Parameter	Return Type
1	getConversations	Returns the list of conversations for the current user, wrapped in a map (e.g. list, pagination info, unread counts).	None	ResponseEntity<Map>
2	createOrGetConversation	Creates a new conversation with the specified user or restores an existing soft-deleted conversation. Requires the current user to follow the target user.	request: Map<String, String> (contains userId)	ResponseEntity<Map>

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3	getConversation Messages	Returns messages for the specified conversation, wrapped in a map (e.g. list, pagination info, unread status).	conversati onId: String	ResponseEntity< Map>
4	sendMessage	Sends a message in the specified conversation. If the users are not mutually following, the current user is limited to a single message in that conversation until followed.	conversati onId: String, request: Map<Strin g,String>	ResponseEntity< Map>
5	markAsRead	Marks a single message as read.	messageId: String	ResponseEntity< Map>
6	markConversation AsRead	Marks all messages in the specified conversation as read for the current user.	conversati onId: String	ResponseEntity< Map>
7	deleteConversatio n	Soft-deletes a conversation from the current user's view without removing the underlying messages or the other user's conversation.	conversati onId: String	ResponseEntity< Map>

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3.2.2.3 Repository

3.2.2.3.1 Repository: `CommunityPostRepository`

<<interface>> CommunityPostRepository	
+save(post: CommunityPost):CommunityPost +findById(id: String):Optional +findAllOrderByCreatedAtDesc():List +findByCommunityOrderByCreatedAtDesc(community: Community):List +findByStatus(status: Status, pageable: Pageable):Page +countByStatus(status: Status):long +findByAuthorIdOrderByCreatedAtDesc(authorId: String):List +findByAuthorIdAndStatusOrderByCreatedAtDesc(authorId: String, status: Status):List +findByTitle(title: String):List	

Methods:

ID	Name	Description	Parameter	Return Type
1	save	Saves a post entity (create or update).	post: CommunityPost	CommunityPost
2	findById	Finds a post by its ID.	id: String	Optional<CommunityPost>
3	findAllOrderByCreatedAtDesc	Finds all posts ordered by creation date descending.	None	List<CommunityPost>
4	findByCommunityOrderByCreatedAtDesc	Finds posts by community ordered by creation date descending.	community: Community	List<CommunityPost>
5	findByStatus	Finds posts by status with pagination.	status: Status, pageable: Pageable	Page<CommunityPost>

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6	countByStatus	Counts posts by status.	status: Status	long
7	findByAuthorId OrderByCreated AtDesc	Finds posts by author ID ordered by creation date descending.	authorId: String	List<CommunityPost>
8	findByAuthorId AndStatusOrder ByCreatedAtDesc	Finds posts by author ID and status ordered by creation date descending.	authorId: String, status: Status	List<CommunityPost>
9	findByTitle	Finds posts by title (for duplicate checking).	title: String	List<CommunityPost>

3.2.2.3.2 Repository: `CommentRepository`

<<interface>> CommentRepository	
+save(comment: Comment):Comment +findByPostOrderByCreatedAtDesc(post: CommunityPost):List +findByPostId(postId: String):List +countByPost(post: CommunityPost):long	

Methods:

ID	Name	Description	Parameter	Return Type
1	save	Saves a comment entity (create or update).	`comment: Comment`	Comment
2	findByPostOrder	Finds comments for a post ordered by creation date descending	`post: CommunityPost`	List<Comment>

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	ByCreatedAtDesc			
3	findByPostId	Finds comments by post ID	`postId: String`	List<Comment>
4	countByPost	Counts comments for a post	`post: CommunityPost`	long

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3.2.2.4 Model

3.2.2.4.1 Model: `CommunityPost`

CommunityPost
+id: String +community: Community +author: AppUser +title: String +body: String +titleZh: String +titleEn: String +contentZh: String +contentEn: String +originalLanguage: String +tags: List +category: String +imageUrl: String +aiResult: String +aiConfidence: Double +status: Status +reviewNote: String +reviewedBy: String +reviewedAt: Instant +embedding: String +createdAt: Instant +updatedAt: Instant +needsManualReview():boolean
+approve(reviewerId: String, note: String):void +reject(reviewerId: String, note: String):void

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Attributes:

ID	Name	Type	Description
1	id	String	Primary key ID (UUID)
2	community	Community	Community this post belongs to (nullable)
3	author	AppUser	Author of the post
4	title	String	Post title
5	body	String	Post content (TEXT)
6	titleZh	String	Chinese translation of title
7	titleEn	String	English translation of title
8	contentZh	String	Chinese translation of content
9	contentEn	String	English translation of content
10	originalLanguage	String	Detected original language (zh/en/th)
11	tags	List<String>	Post tags (e.g., Study, Housing, Travel)
12	category	String	Post category
13	imageUrl	String	Image URL if post contains images
14	aiResult	String	AI moderation result (TEXT)
15	aiConfidence	Double	AI confidence score (0.0 - 1.0)

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16	status	Status	Post moderation status (PENDING_REVIEW, APPROVED, REJECTED, REPORTED_REMOVED)
17	reviewNote	String	Review note (filled by AI system with reviewerId "AI_AUTOMOD")
18	reviewedBy	String	Reviewer ID ("AI_AUTOMOD" for AI-approved/rejected posts)
19	reviewedAt	Instant	Review timestamp
20	embedding	String	Semantic embedding for search (TEXT)
21	createdAt	Instant	Time when the post was created
22	updatedAt	Instant	Time when the post was last updated

Methods:

ID	Name	Description	Parameter	Return Type
1	needsManualReview	Indicates whether the post requires additional AI-based safety checks (no manual review).	None	boolean

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2	approve	Approves the post with reviewer information	`reviewer Id: String`, `note: String`	void
3	reject	Rejects the post with reviewer information	`reviewer Id: String`, `note: String`	void

3.2.2.4.2 Model: `Comment`

Comment
+id: String +post: CommunityPost +author: AppUser +content: String +contentZh: String +contentEn: String +originalLanguage: String +status: Status +createdAt: Instant +updatedAt: Instant

Attributes:

ID	Name	Type	Description
1	id	String	Primary key ID (UUID)

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2	post	CommunityPost	Post this comment belongs to
3	author	AppUser	Author of the comment
4	content	String	Comment content (TEXT)
5	contentZh	String	Chinese translation of content
6	contentEn	String	English translation of content
7	originalLanguage	String	Detected original language (zh/en/th)
8	status	Status	Comment status (ACTIVE, REPORTED_REMOVED)
9	createdAt	Instant	Time when the comment was created
10	updatedAt	Instant	Time when the comment was last updated

3.2.2.4.3 Model: `AppUser`

AppUser
+id: String +username: String +displayName: String +email: String +avatar: String

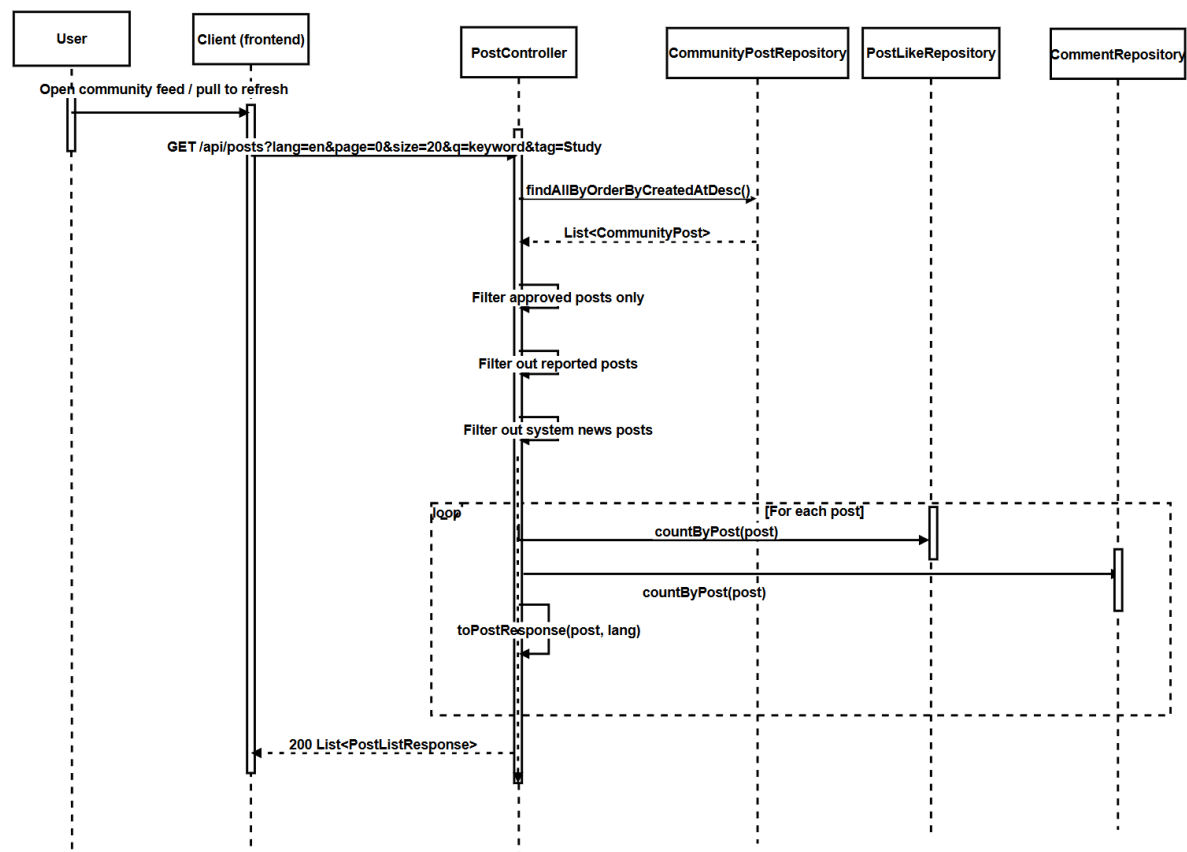
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Attributes:

ID	Name	Description	Type
1	Id	Primary key ID	String
2	Username	Username for login	String
3	Display	Display name shown to other users	String
4	Email	User email address	String
5	Avatar	User avatar URL	String

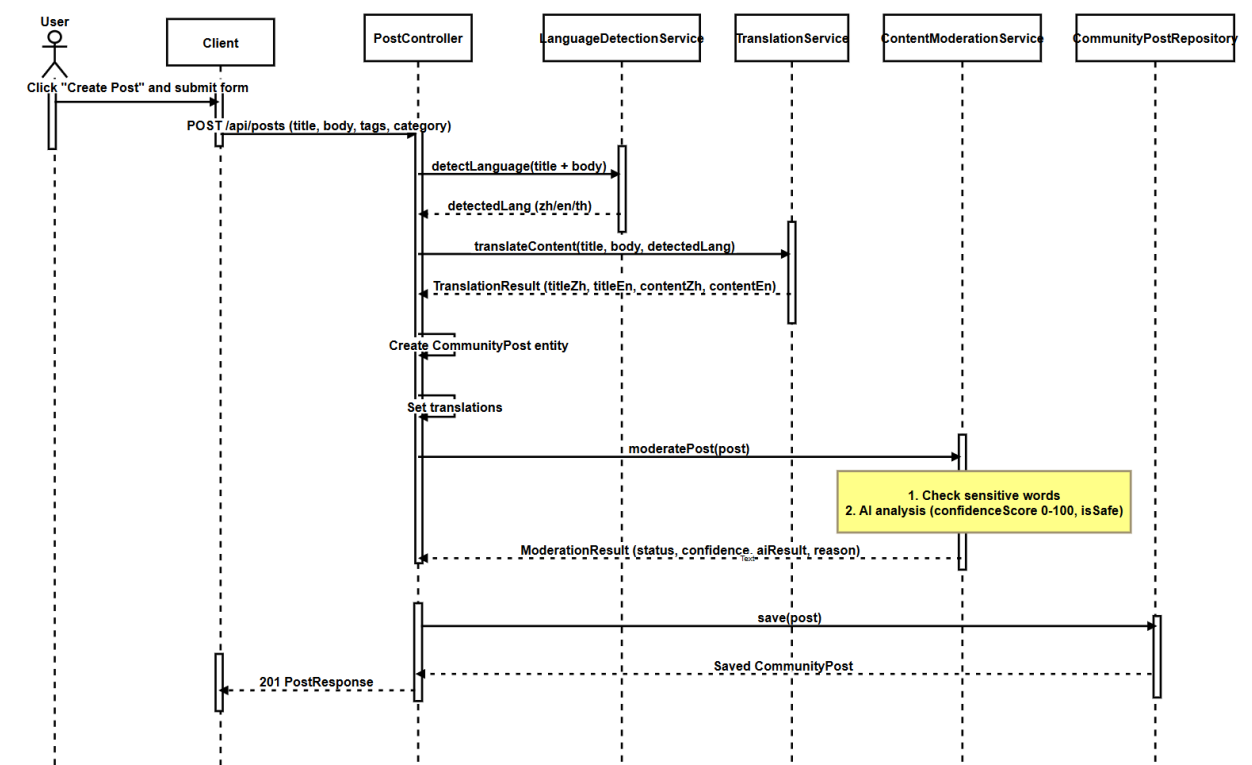
3.2.3 Sequence Diagram

SD-07: Browse Community Feed



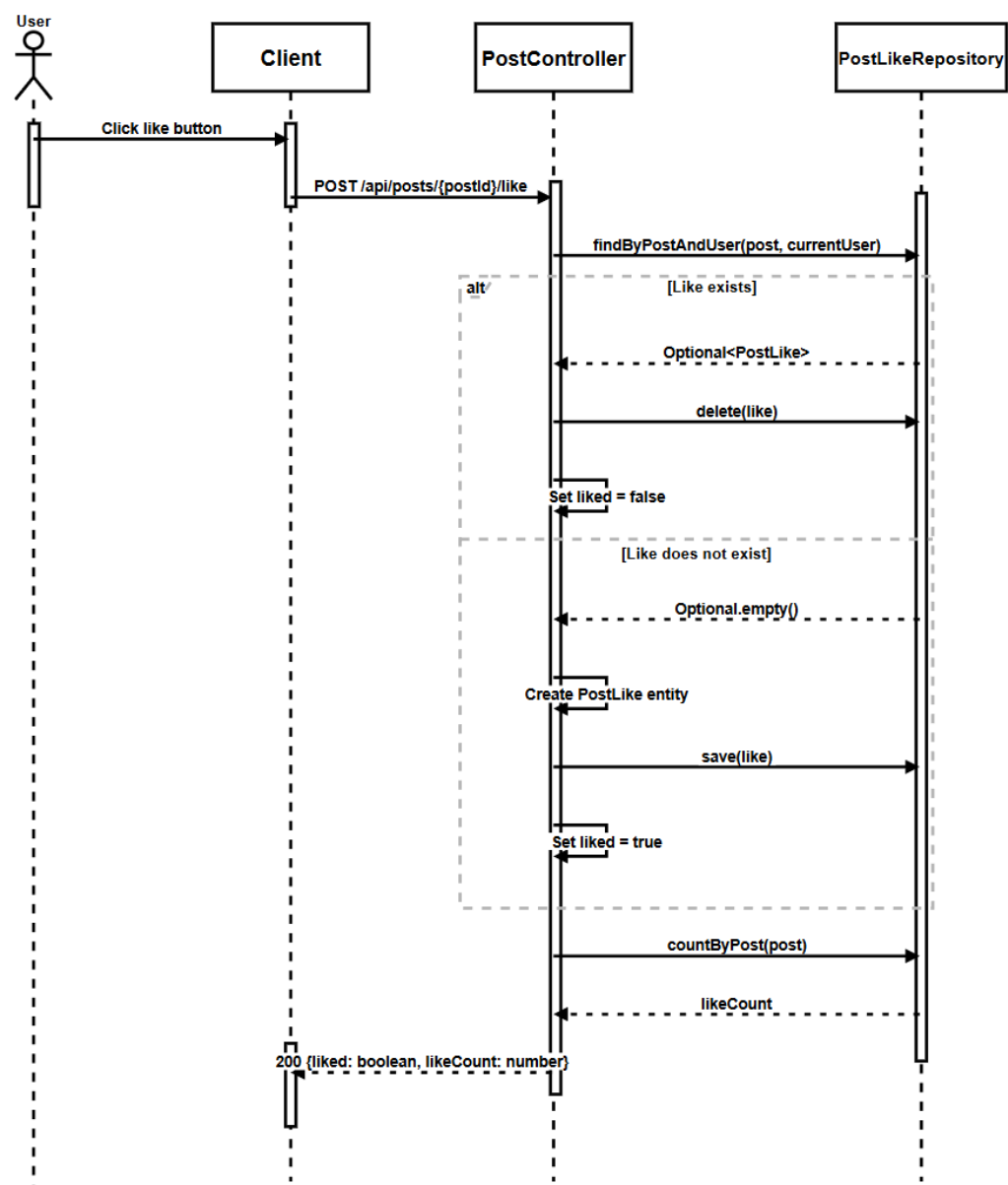
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SD-08: Create Publish Post



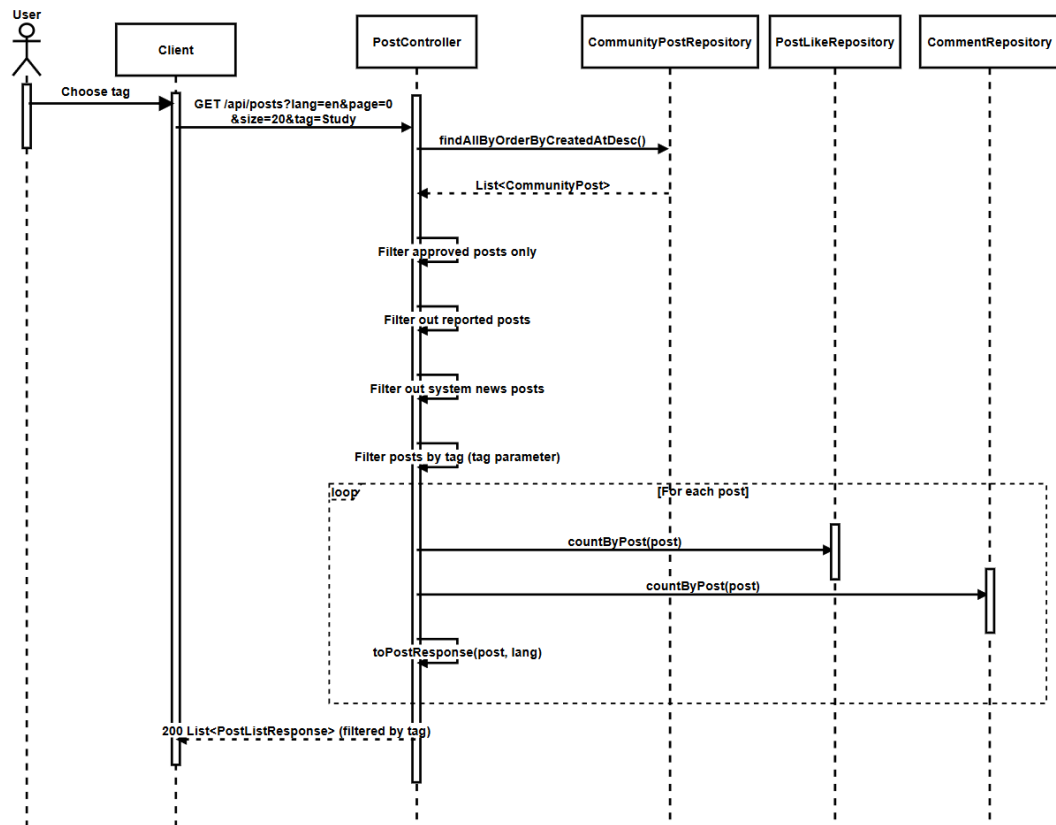
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	48
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SD-09:Like Post



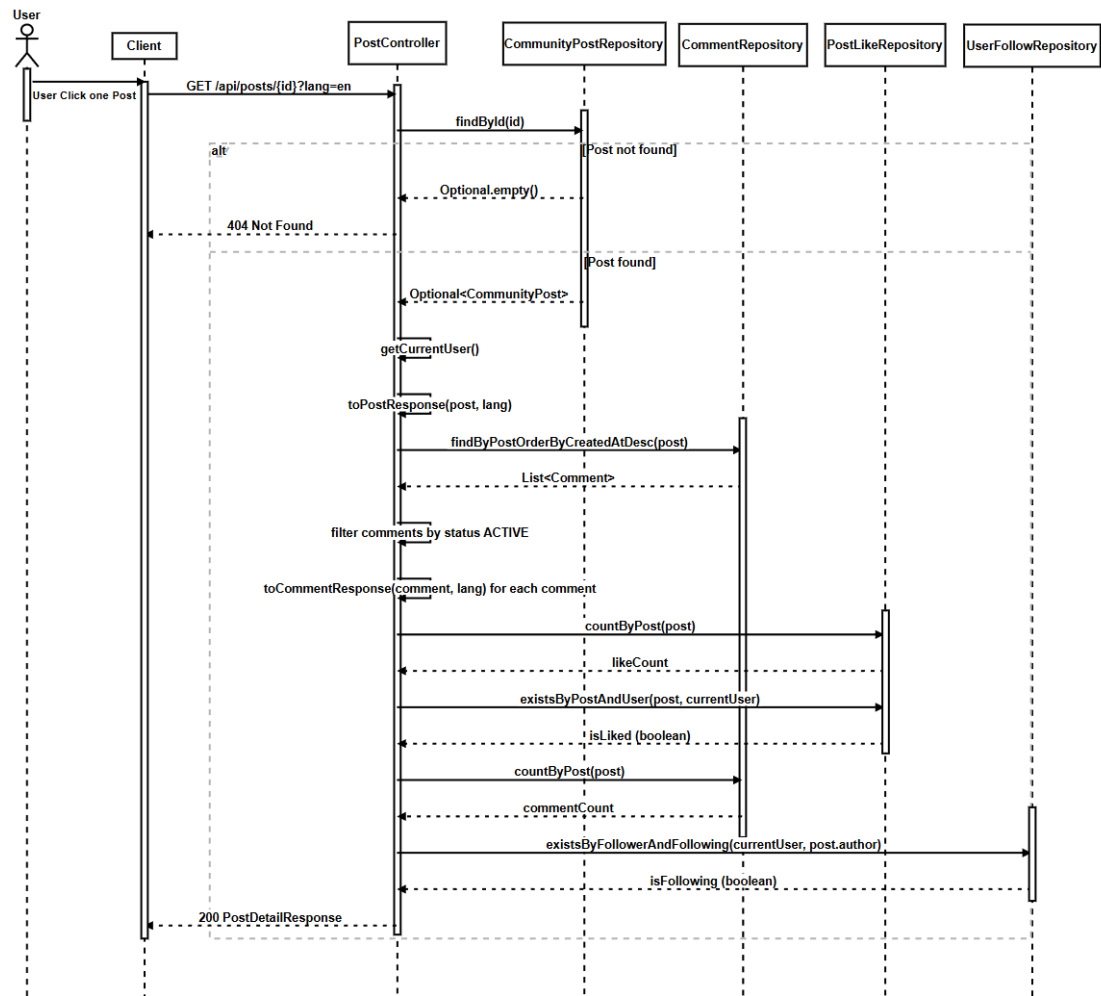
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SD-10:Filter Post by tag



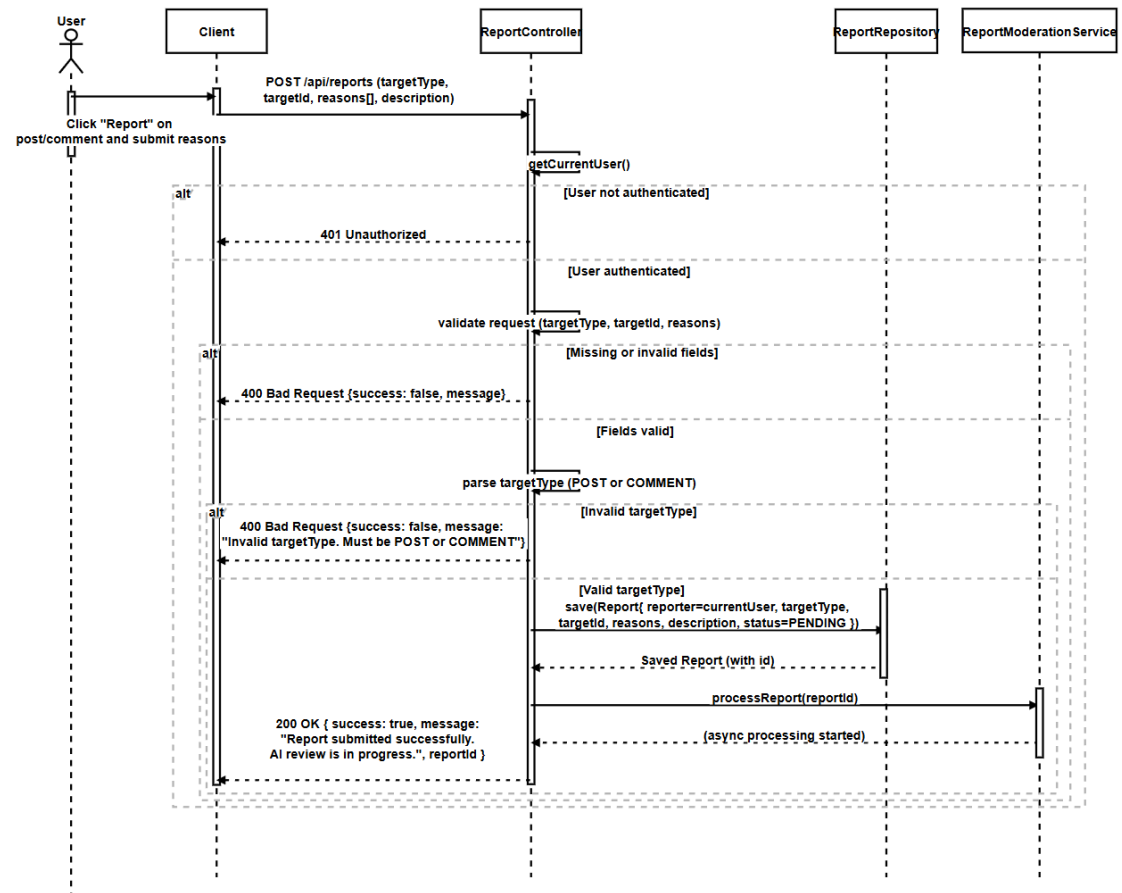
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SD-11: View Post Details



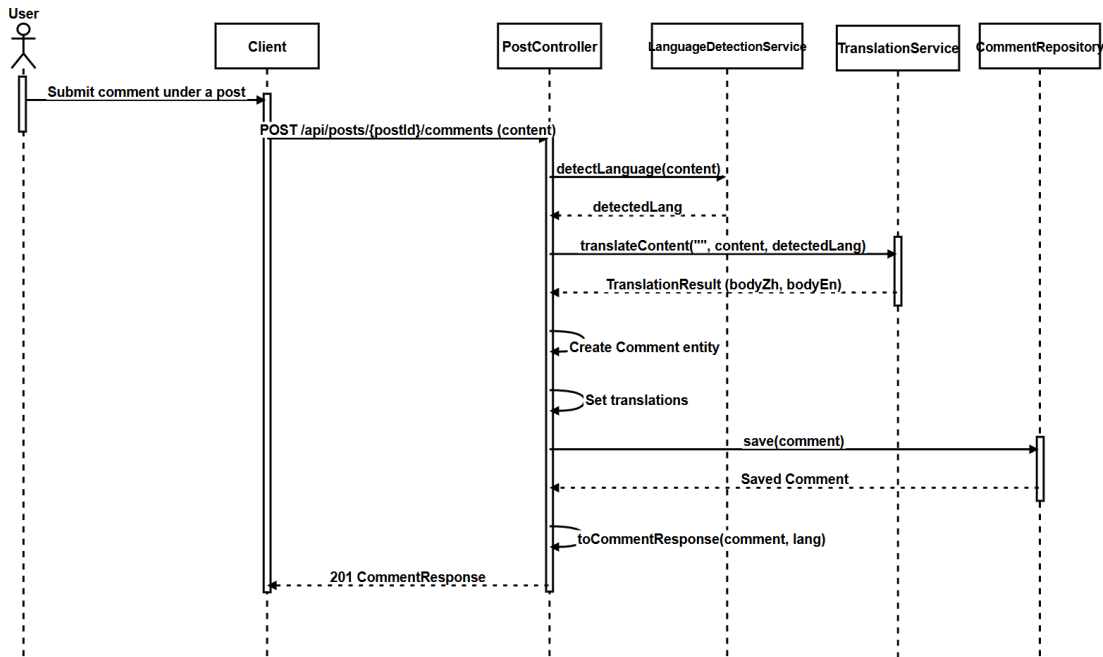
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	51
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SD-12: Report Post/Comment



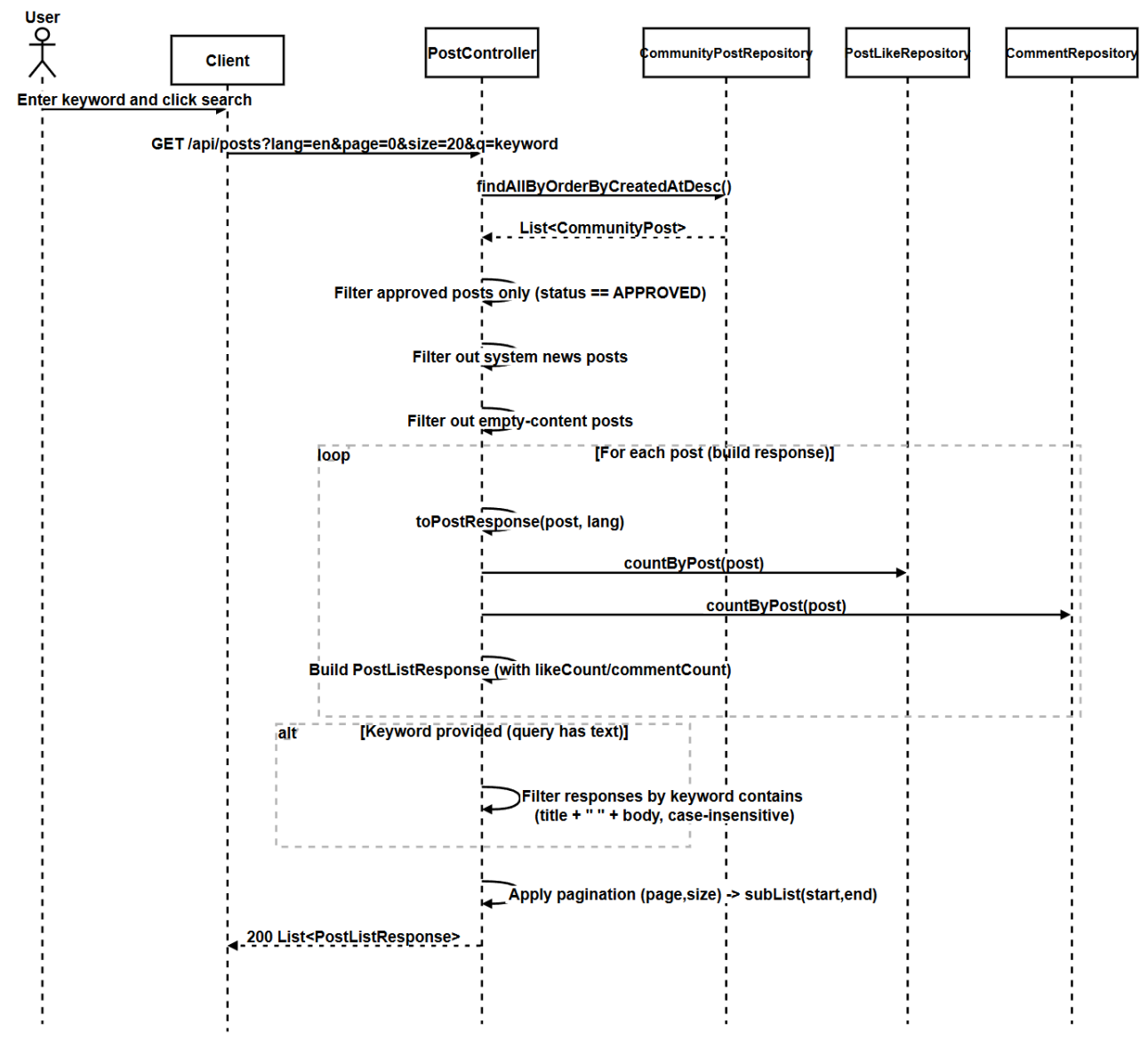
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	52
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SD-13: Post Comment



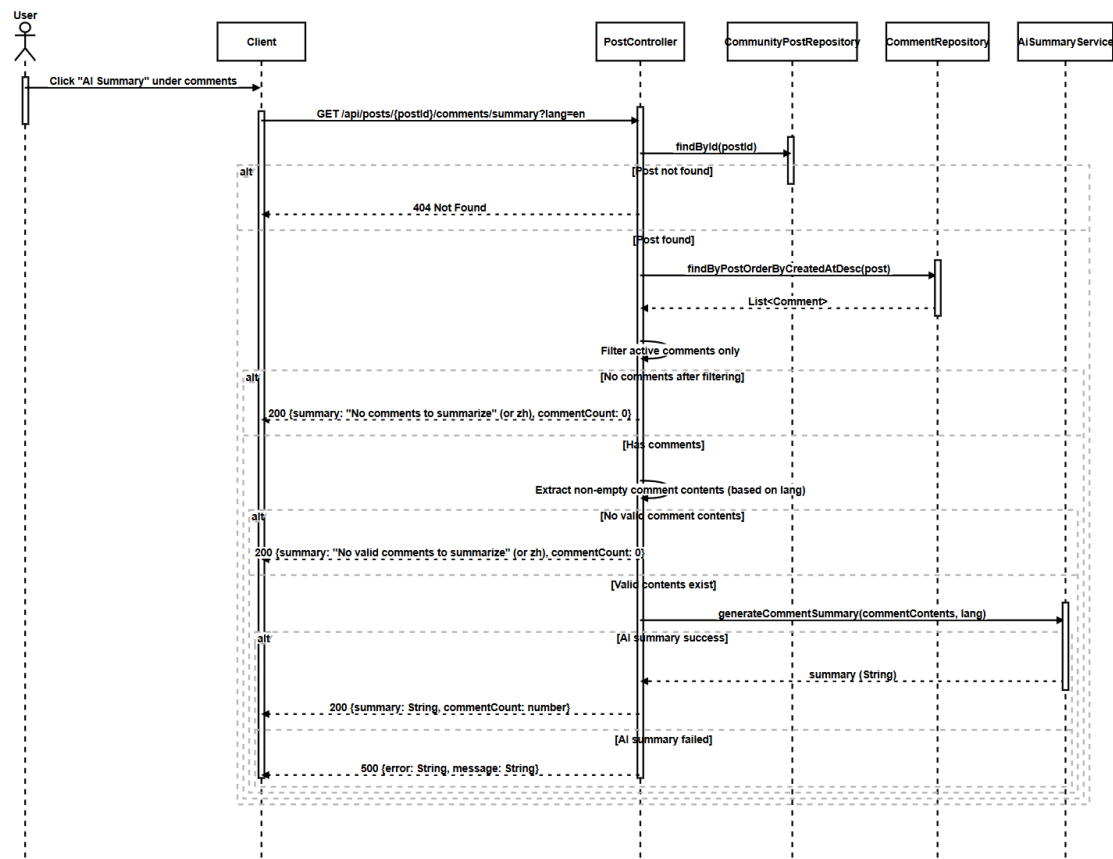
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	53
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SD-14: Search Posts



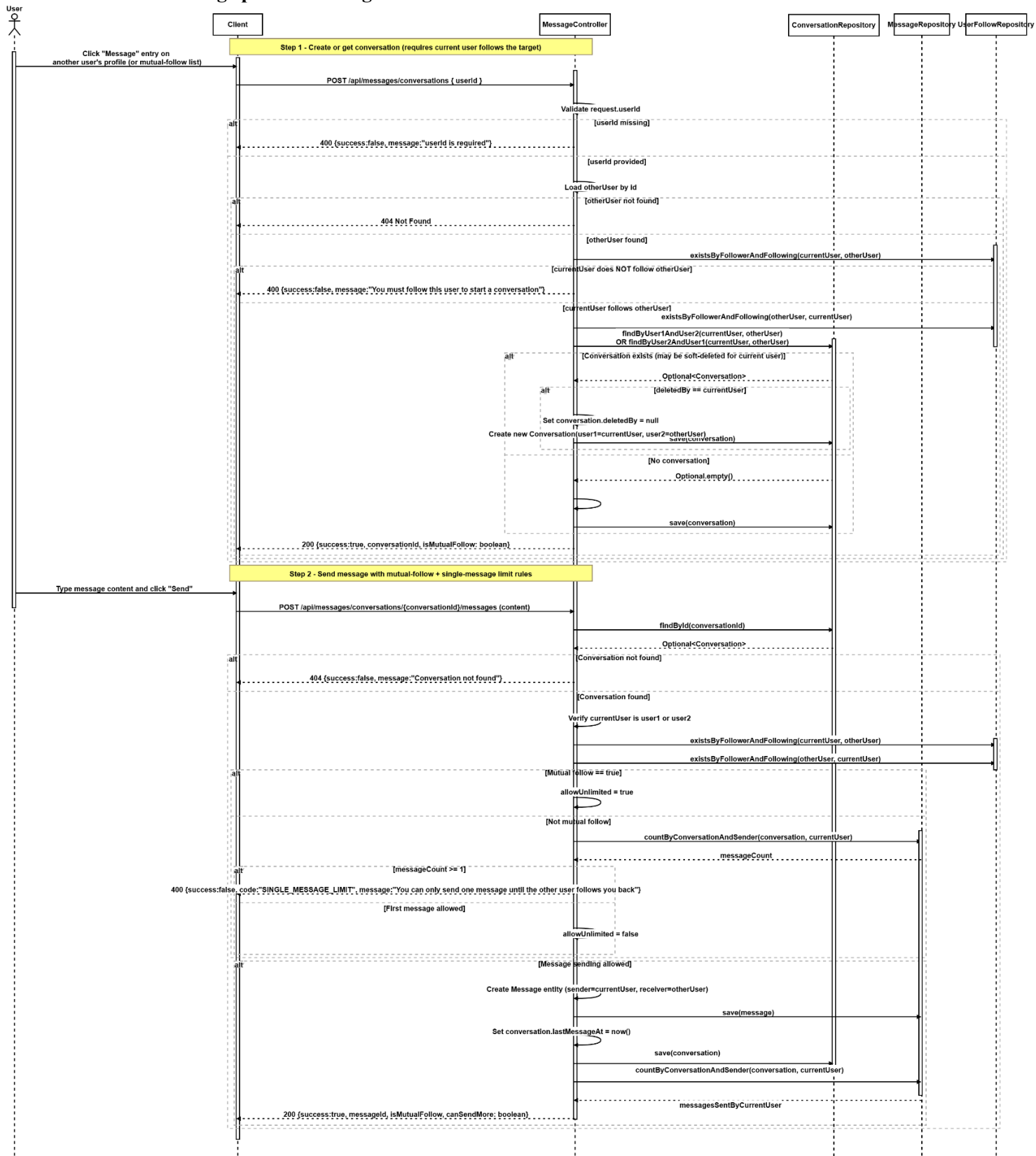
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	54
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SD-15: AI Summary Comment



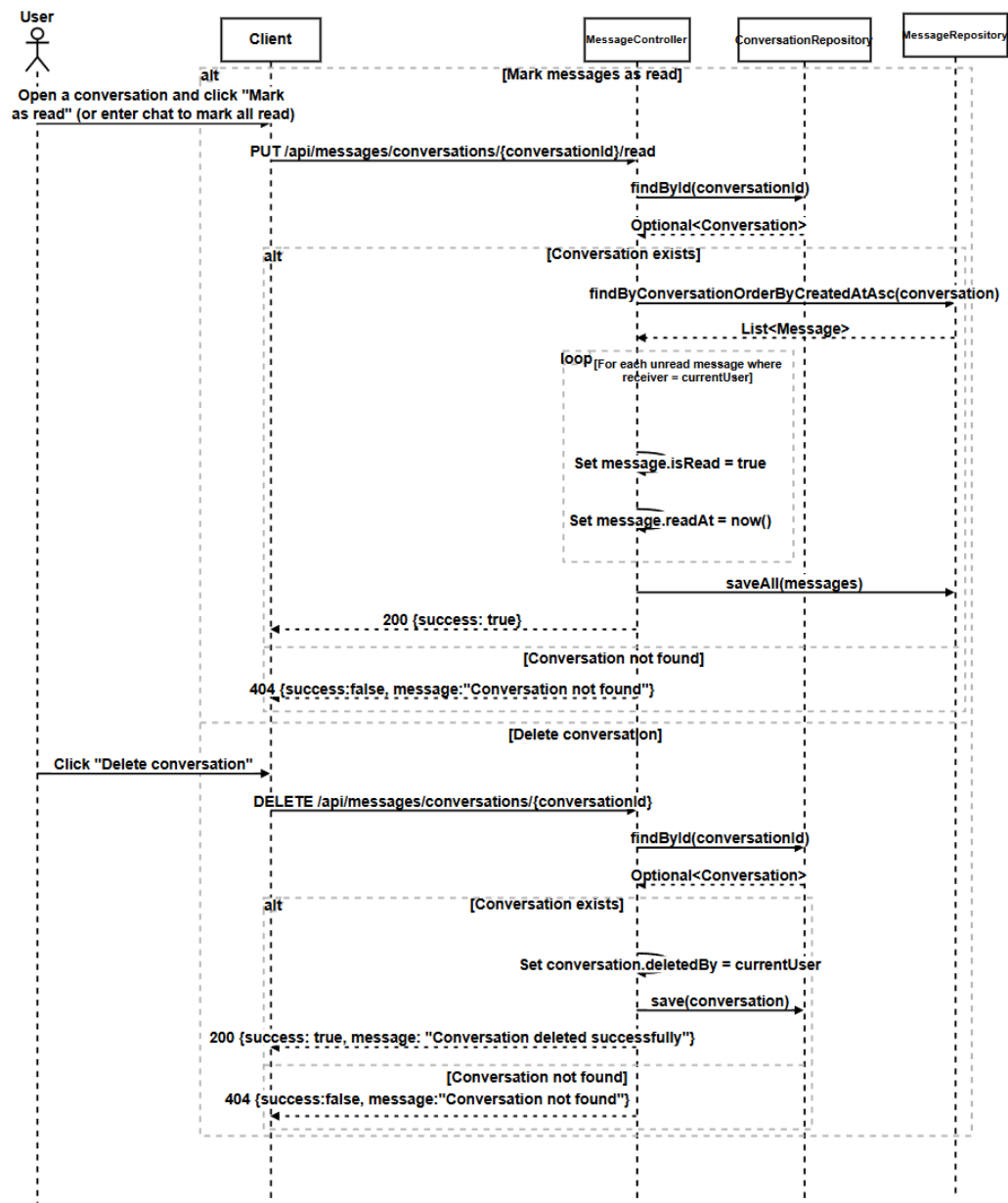
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	55
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SD-16:Manage private messages with other users



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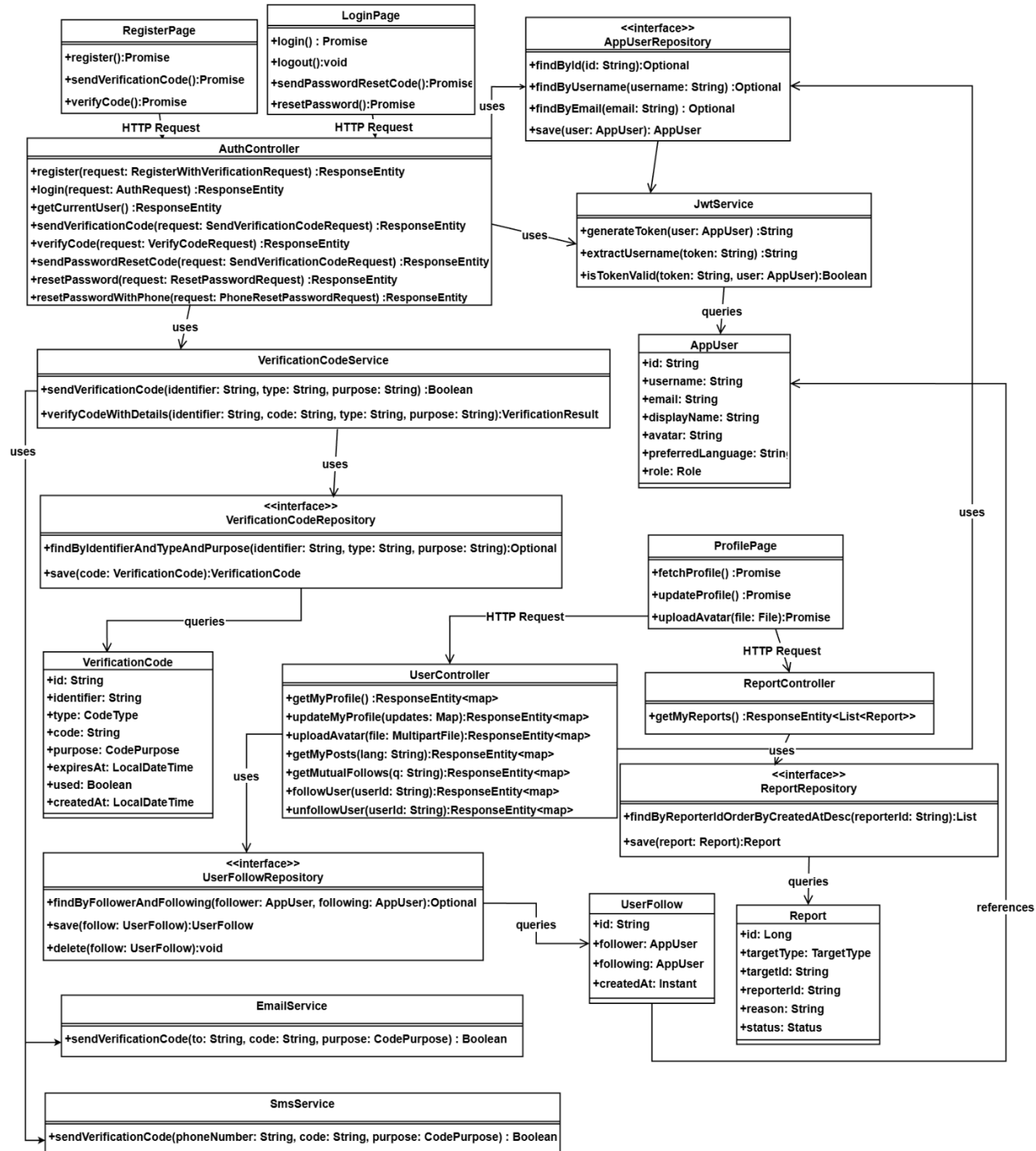
SD-17: Manage Message Status



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3.3 Feature 3

3.3.1 Class Diagram

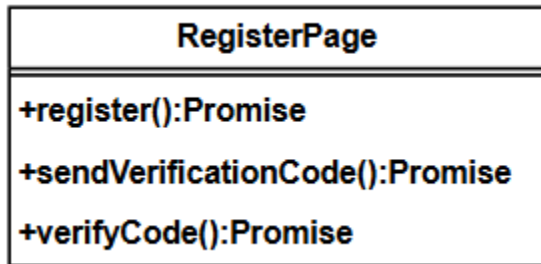


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3.3.2 Class Diagram Description

3.3.2.1 View

3.3.2.1.1 RegisterPage



Methods:

ID	Name	Description	Parameter	Return Type
1	register	Registers a new user account	None	Promise<void>
2	sendVerificationCode	Sends a verification code for registration	None	Promise<void>
3	verifyCode	Verifies the registration verification code	None	Promise<void>

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3.3.2.1.2 LoginPage

LoginPage
+login() : Promise +logout():void +sendPasswordResetCode():Promise +resetPassword():Promise

Methods:

ID	Name	Description	Parameters	Return Type
1	login	Authenticates user and logs in	None	Promise<void>
2	logout	Logs out the current user	None	void
3	sendPasswordResetCode	Sends password reset verification code (email or phone)	None	Promise<void>
4	resetPassword	Resets password using verification code	None	Promise<void>

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3.3.2.1.3 ProfilePage

ProfilePage
+fetchProfile() :Promise +updateProfile() :Promise +uploadAvatar(file: File):Promise

Methods:

ID	Name	Description	Parameters	Return Type
1	fetchProfile	Fetches current user's profile information	None	Promise<void>
2	updateProfile	Updates user's profile information	None	Promise<void>
3	uploadAvatar	Uploads a new avatar image for the user	file: File	Promise<void>

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3.3.2.2 Controller

3.3.2.2.1 Controller: `AuthController`

AuthController
<pre>+register(request: RegisterWithVerificationRequest) :ResponseEntity +login(request: AuthRequest) :ResponseEntity +getCurrentUser() :ResponseEntity +sendVerificationCode(request: SendVerificationCodeRequest) :ResponseEntity +verifyCode(request: VerifyCodeRequest) :ResponseEntity +sendPasswordResetCode(request: SendVerificationCodeRequest) :ResponseEntity +resetPassword(request: ResetPasswordRequest) :ResponseEntity +resetPasswordWithPhone(request: PhoneResetPasswordRequest) :ResponseEntity</pre>

Methods:

ID	Name	Description	Parameters	Return Type
1	register	Registers a new user account	request: Register WithVerif icationRe quest	ResponseEntity <AuthResponse >
2	login	Authenticates user and returns JWT token	request: AuthRequ est	ResponseEntity <AuthResponse >
3	getCurrentUs er	Retrieves current authenticated user information	None	ResponseEntity <UserDTO>
4	sendVerificat ionCode	Sends verification code to email or phone	request: SendVerif icationCo	ResponseEntity

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			deReques t	
5	verifyCode	Verifies verification code	request: VerifyCo deReques t	ResponseEntity
6	sendPasswor dResetCode	Sends password reset verification code to email or phone	request: SendVerif icationCo deReques t	ResponseEntity
7	resetPasswor d	Resets password using email verification code	request: ResetPass wordReq uest	ResponseEntity
8	resetPasswor dWithPhone	Resets password using phone (Firebase verified)	request: PhoneRes etPasswor dRequest	ResponseEntity

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3.3.2.2.2 Controller: `UserController`

UserController	
+getMyProfile() :ResponseEntity<map> +updateMyProfile(updates: Map):ResponseEntity<map> +uploadAvatar(file: MultipartFile):ResponseEntity<map> +getMyPosts(lang: String):ResponseEntity<map> +getMutualFollows(q: String):ResponseEntity<map> +followUser(userId: String):ResponseEntity<map> +unfollowUser(userId: String):ResponseEntity<map>	

Methods:

ID	Name	Description	Parameters	Return Type
1	getMyProfile	Retrieves current user's profile information	None	ResponseEntity<Map>
2	updateMyProfile	Updates current user's profile	updates: Map	ResponseEntity<Map>
3	uploadAvatar	Uploads avatar image for current user	file: MultipartFile	ResponseEntity<Map>
4	getMyPosts	Retrieves current user's posts	lang: String	ResponseEntity<Map>
5	getMutualFollows	Retrieves mutual follows list for current user	q: String	ResponseEntity<Map>

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6	followUser	Follows a user	userId: String	ResponseEntity <Map>
7	unfollowUser	Unfollows a user	userId: String	ResponseEntity <Map>

3.3.2.2.3 Controller: `ReportController`

ReportController
+getMyReports() :ResponseEntity<List<Report>>

Methods:

ID	Name	Description	Parameter	Return Type
1	getMyReports	Retrieves all reports submitted by the current user	None	`ResponseEntity<List<Report>>`

3.3.2.3 Repository

3.3.2.3.1 Repository: `AppUserRepository`

<<interface>> AppUserRepository
+findById(id: String):Optional +findByUsername(username: String) :Optional +findByEmail(email: String) : Optional +save(user: AppUser): AppUser

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Methods:

ID	Name	Description	Parameters	Return Type
1	findById	Finds a user by ID	id: String	Optional<App User>
2	findByUsername	Finds a user by username	username: String	Optional<App User>
3	findByEmail	Finds a user by email	email: String	Optional<App User>
4	save	Saves a user entity	user: AppUser	AppUser

3.3.2.3.2 Repository: `UserFollowRepository`

<<interface>> UserFollowRepository	
+findByFollowerAndFollowing(follower: AppUser, following: AppUser):Optional +save(follow: UserFollow):UserFollow +delete(follow: UserFollow):void	

Methods:

ID	Name	Description	Parameters	Return Type
1	findByFollowerAndFollowing	Finds a follow relationship by follower and following users	follower: AppUser, following: AppUser	Optional<User Follow>

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2	save	Saves a follow relationship entity	follow: UserFollow	UserFollow
3	delete	Deletes a follow relationship	follow: UserFollow	void

3.3.2.3.3 Repository: `ReportRepository`

<<interface>> ReportRepository	
+findByReporterIdOrderByCreatedAtDesc(reporterId: String):List +save(report: Report):Report	

Methods:

ID	Name	Description	Parameter	Return Type
1	findByReporterIdOrderByCreatedAtDesc	Finds all reports submitted by a specific reporter, ordered by creation time descending	`reporterId: String`	List<Report>
2	save	Saves a report entity	`report: Report`	Report

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3.3.2.3.4 Repository: `VerificationCodeRepository`

<<interface>> VerificationCodeRepository	
+findByIdentifierAndTypeAndPurpose(identifier: String, type: String, purpose: String):Optional	
+save(code: VerificationCode):VerificationCode	

Methods:

ID	Name	Description	Parameter	Return Type
1	findByIdentifierAndTypeAndPurpose	Finds a verification code by identifier, type, and purpose	`identifier: String`, `type: String`, `purpose: String`	Optional<VerificationCode>
2	save	Saves a verification code entity	`code: VerificationCode`	VerificationCode

3.3.2.4 Model

3.3.2.4.1 Model: `AppUser`

AppUser
+id: String
+username: String
+email: String
+displayName: String
+avatar: String
+preferredLanguage: String
+role: Role

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Attributes:

ID	Name	Type	Description
1	id	String	Primary key ID
2	username	String	Username
3	email	String	Email address
4	displayName	String	Display name
5	avatar	String	Avatar image URL
6	preferredLanguage	String	Preferred interface language
7	role	Role	User role

3.3.2.4.2 Model: `UserFollow`

UserFollow
+id: String
+follower: AppUser
+following: AppUser
+createdAt: Instant

Attributes:

ID	Name	Type	Description
1	id	String	Primary key ID

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2	follower	AppUser	User who is following
3	following	AppUser	User being followed
4	createdAt	Instant	When the follow relationship was created

3.3.2.4.3 Model: `Report`

Report
+id: Long +targetType: TargetType +targetId: String +reporterId: String +reason: String +status: Status

Attributes:

ID	Name	Type	Description
1	id	Long	Primary key ID
2	targetType	TargetType	Type of target being reported
3	targetId	String	ID of the target being reported
4	reporterId	String	ID of the user who submitted the report

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5	reason	String	Reason for the report
6	status	Status	Status of the report

3.3.2.4.4 Model: `VerificationCode`

VerificationCode
+id: String +identifier: String +type: CodeType +code: String +purpose: CodePurpose +expiresAt: LocalDateTime +used: Boolean +createdAt: LocalDateTime

Attributes:

ID	Name	Type	Description
1	id	String	Primary key ID
2	identifier	String	Email address or phone number
3	type	CodeType	Verification code type (EMAIL or SMS)
4	code	String	Verification code (6 digits)

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5	purpose	CodePurpose	Code purpose (REGISTER or RESET_PASSWORD)
6	expiresAt	LocalDateTime	Expiration time
7	used	Boolean	Whether the code has been used
8	createdAt	LocalDateTime	Creation time

3.3.2.5 Service

3.3.2.5.1 Service: `JwtService`

JwtService	
+generateToken(user: AppUser) :String +extractUsername(token: String) :String +isTokenValid(token: String, user: AppUser):Boolean	

Methods:

ID	Name	Description	Parameters	Return Type
1	generateToken	Generates a JWT token for the given user	user: AppUser	String
2	extractUsername	Extracts username from JWT token	token: String	String
3	isTokenValid	Validates if the JWT token is valid for the given user	token: String,	Boolean

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			user: AppUser	
--	--	--	------------------	--

3.3.2.5.2 Service: `VerificationCodeService`

VerificationCodeService
+sendVerificationCode(identifier: String, type: String, purpose: String) :Boolean
+verifyCodeWithDetails(identifier: String, code: String, type: String, purpose: String):VerificationResult

Methods:

ID	Name	Description	Parameter	Return Type
1	sendVerificationCode	Sends a verification code to the specified identifier	`identifier: String`, `type: String`, `purpose: String`	Boolean
2	verifyCodeWithDetails	Verifies a verification code with detailed result	`identifier: String`, `code: String`, `type: String`, `purpose: String`	VerificationResult

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3.3.2.5.3 Service: `EmailService`

EmailService
+sendVerificationCode(to: String, code: String, purpose: CodePurpose) : Boolean

Methods:

ID	Name	Description	Parameter	Return Type
1	sendVerificationCode	Sends a verification code email to the specified email address	`to: String`, `code: String`, `purpose: CodePurpose`	Boolean

3.3.2.5.4 Service: SMSService

SmsService
+sendVerificationCode(phoneNumber: String, code: String, purpose: CodePurpose) : Boolean

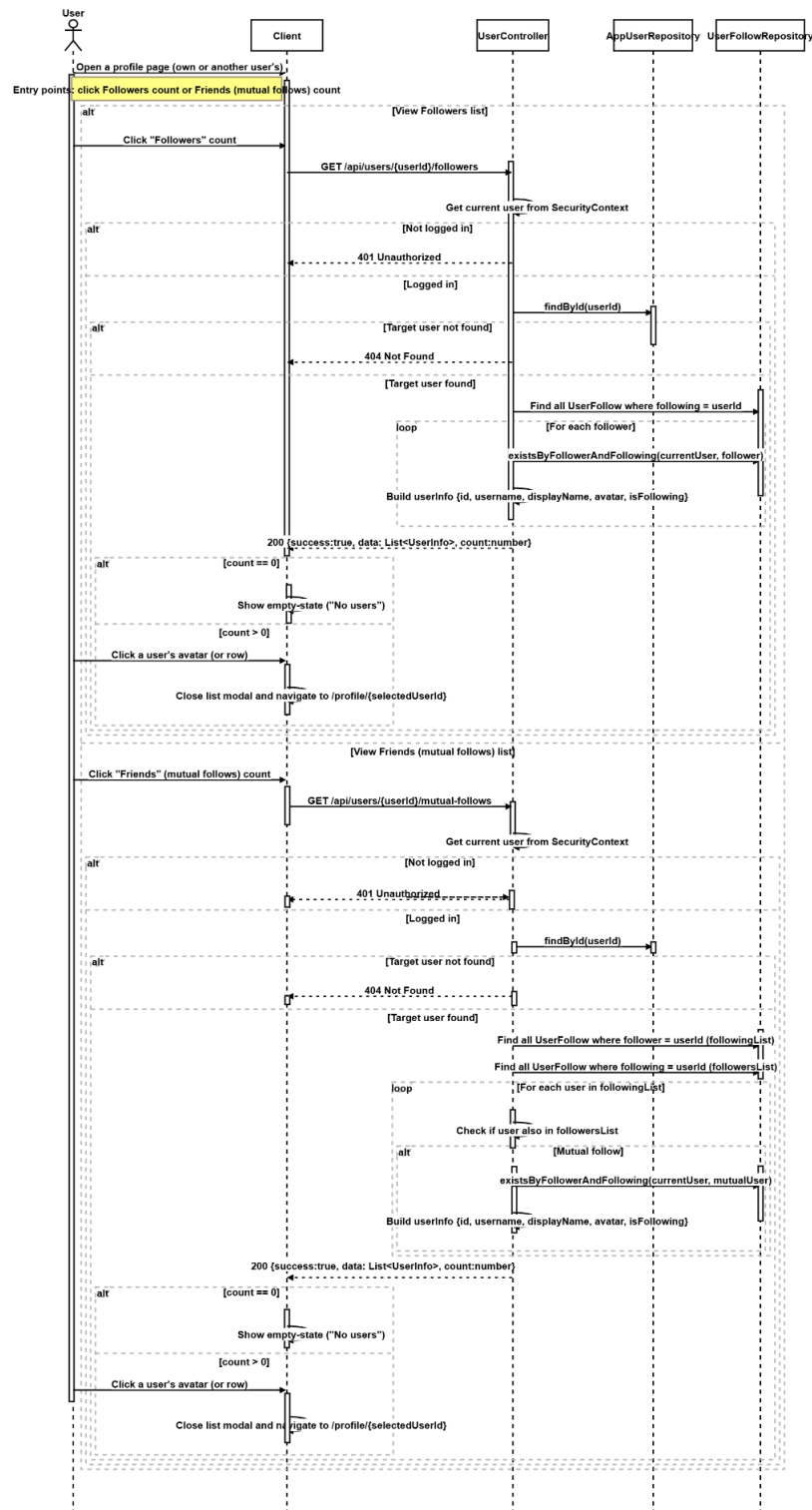
Methods:

ID	Name	Description	Parameter	Return Type
1	sendVerificationCode	Sends a verification code SMS to the specified phone number	`phoneNumber : String`, `code: String`, `purpose: CodePurpose`	Boolean

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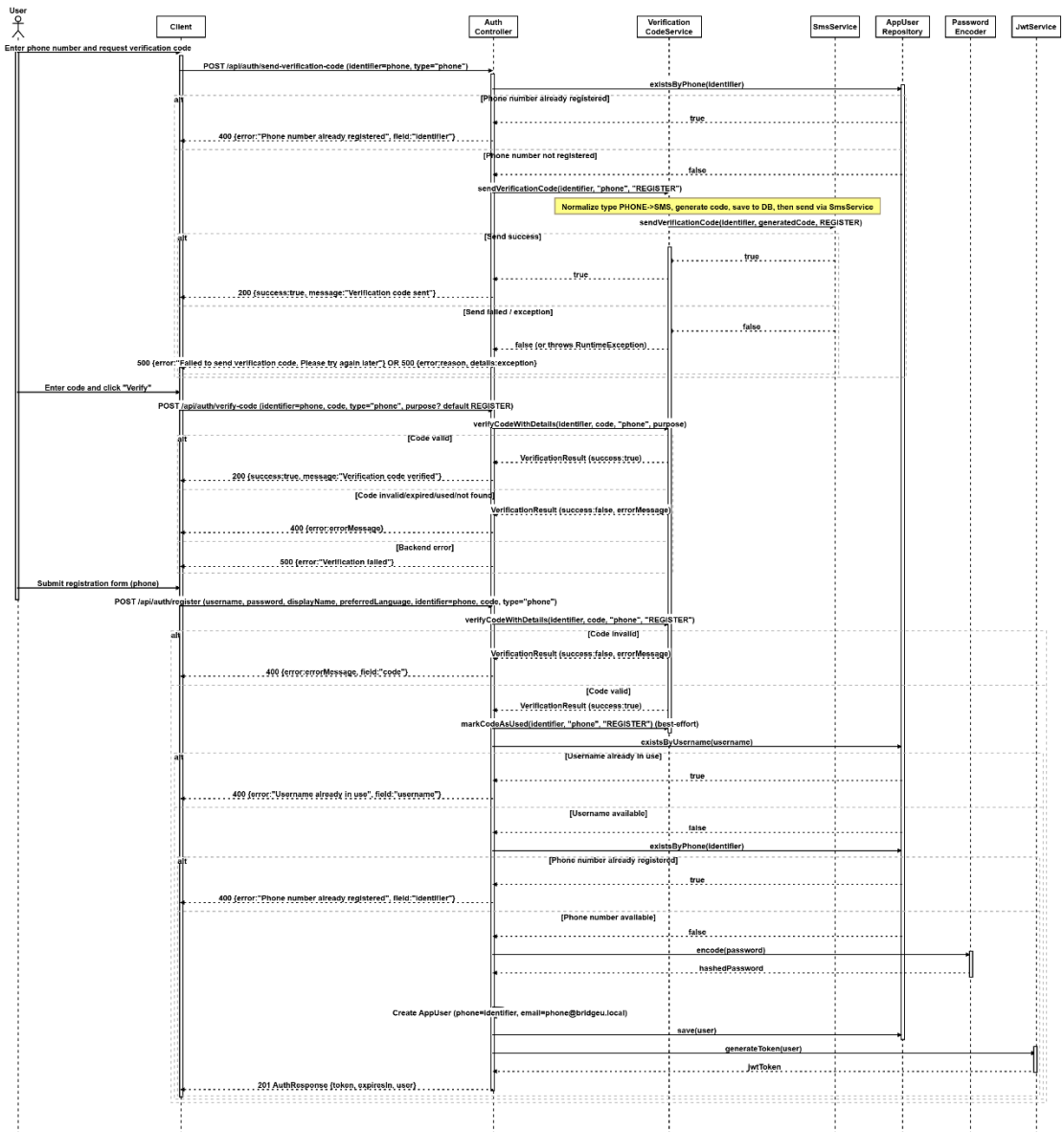
3.3.3 Sequence Diagram

SD-18:View the Friends/Followers list



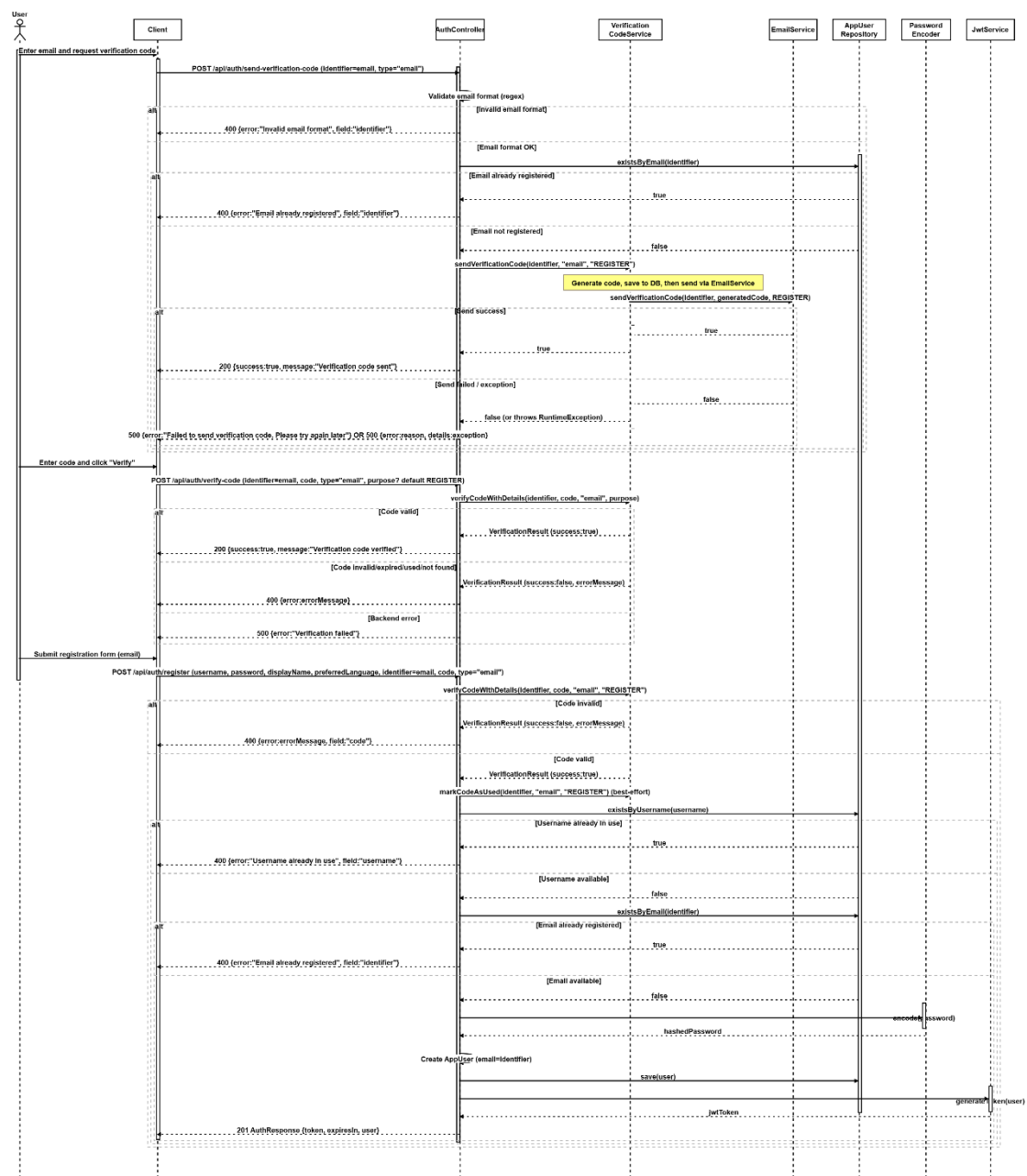
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	75
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SD-19: Register Account (Phone)



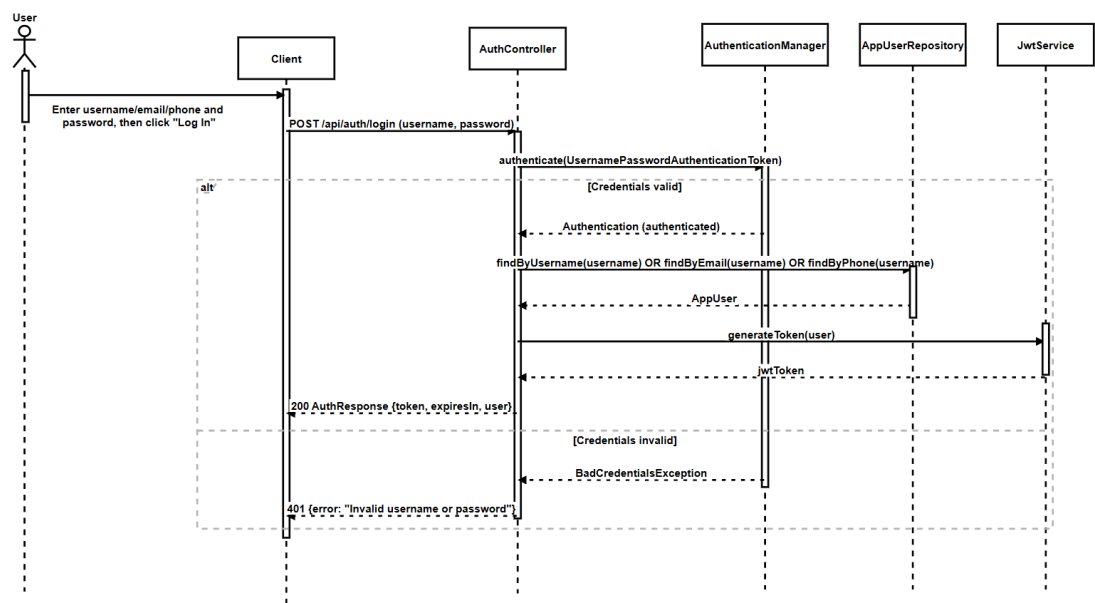
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SD-20: Register Account (Email)

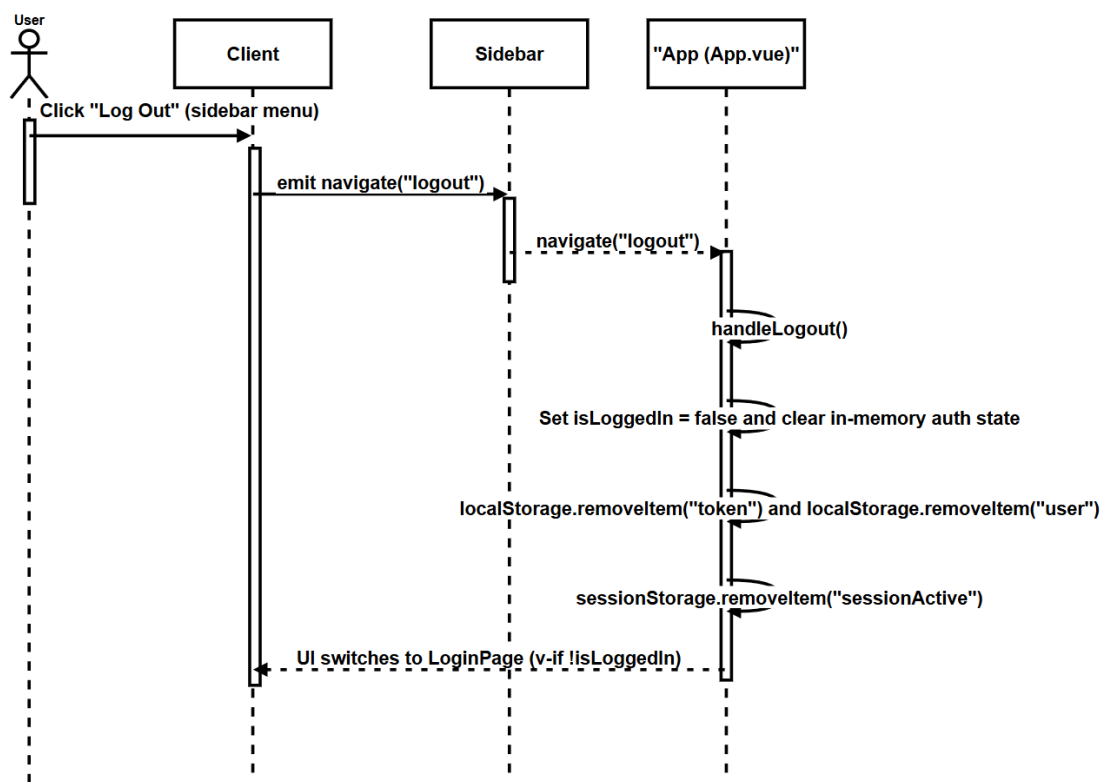


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SD-21: Log In

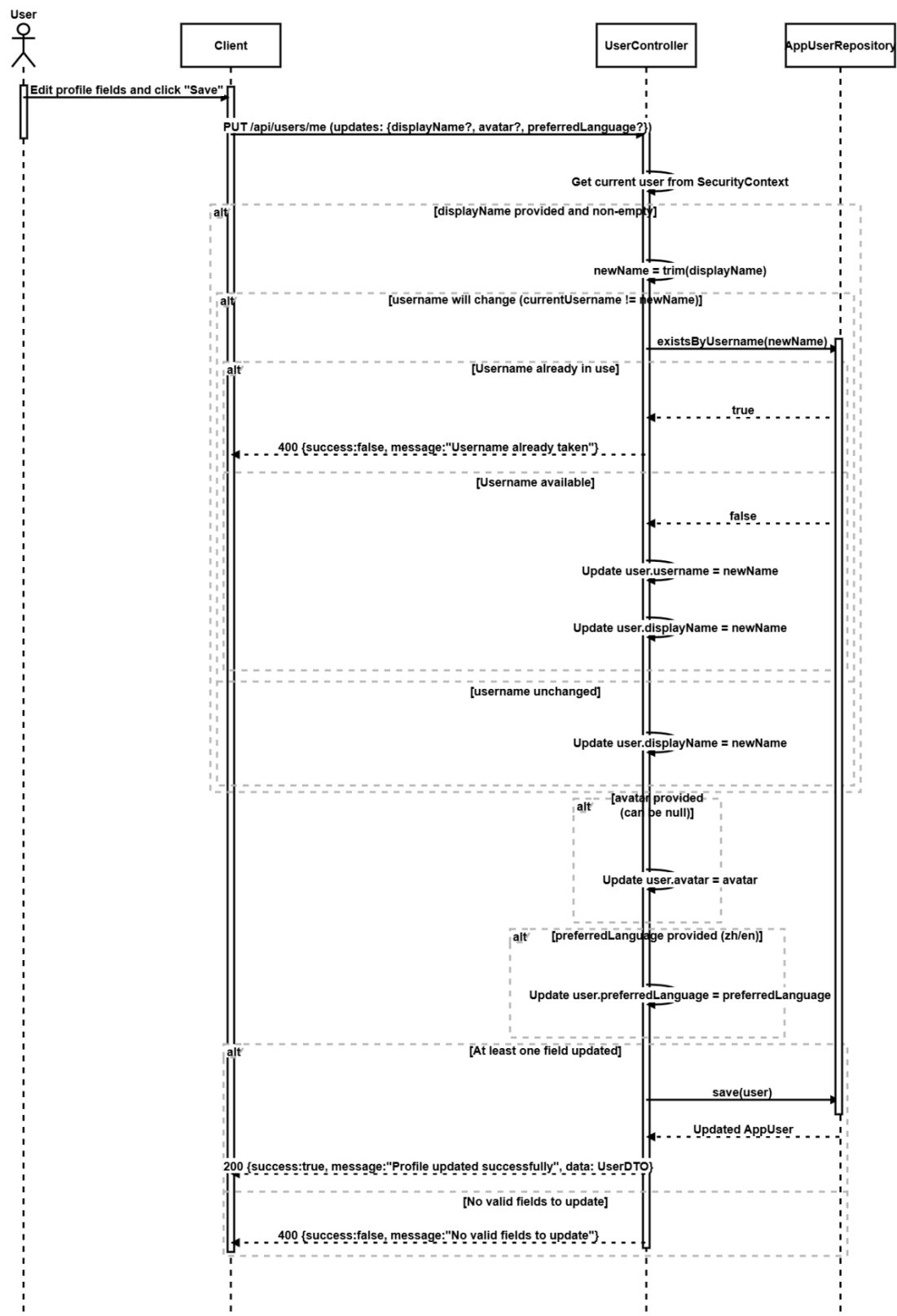


SD-22:Log Out



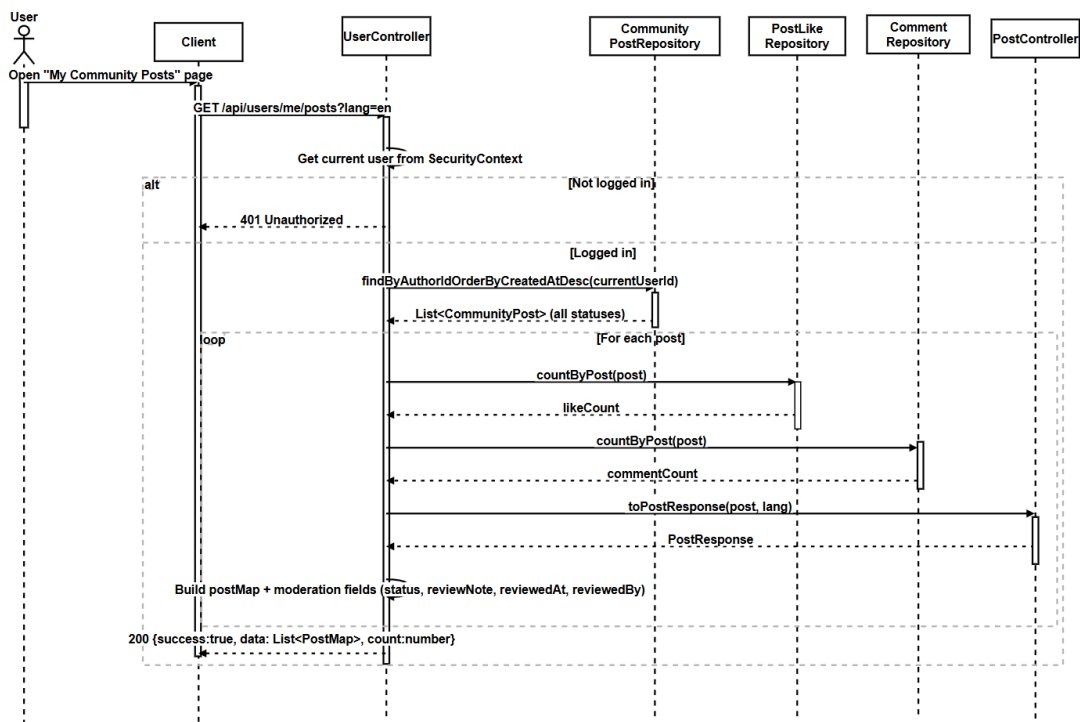
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SD-23: Edit Profile

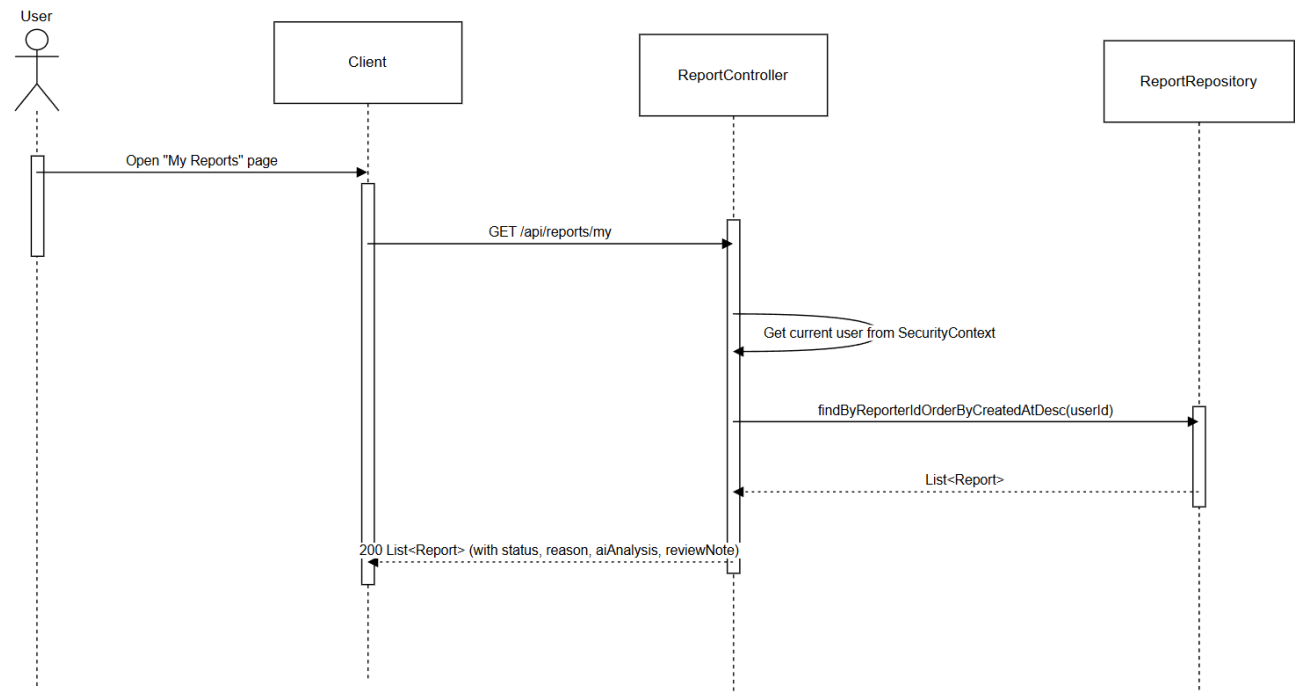


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SD-24: View My Community Posts

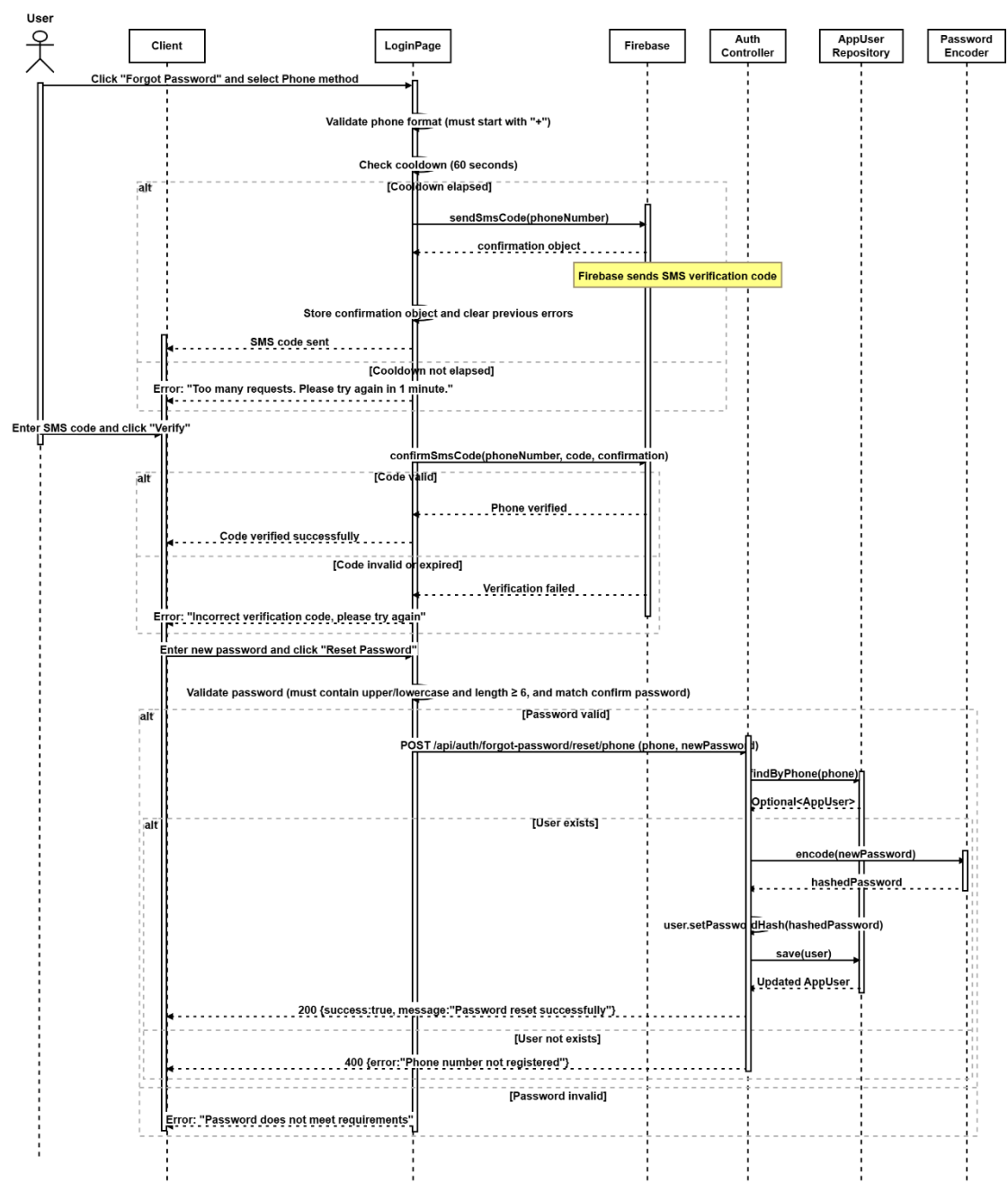


SD-25: View My Reports



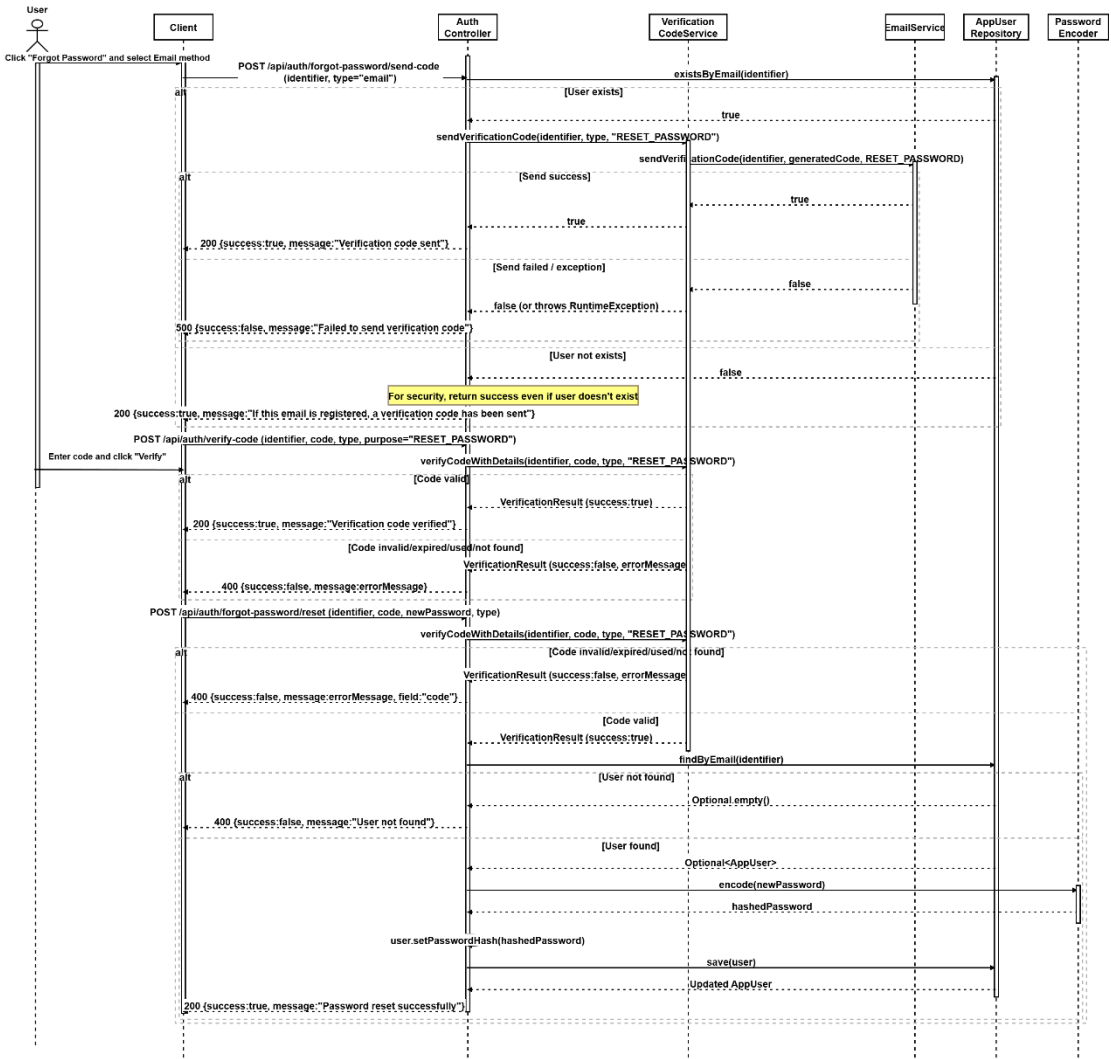
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SD-26: SMS-based Password Reset



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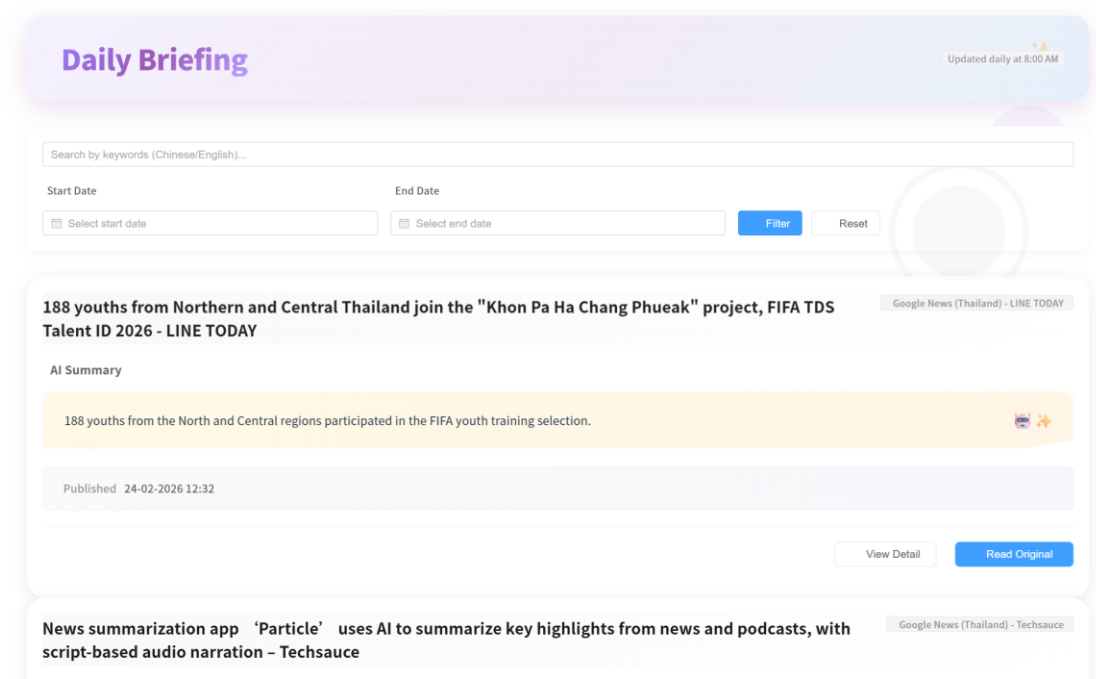
SD-27: Email-based Password Reset



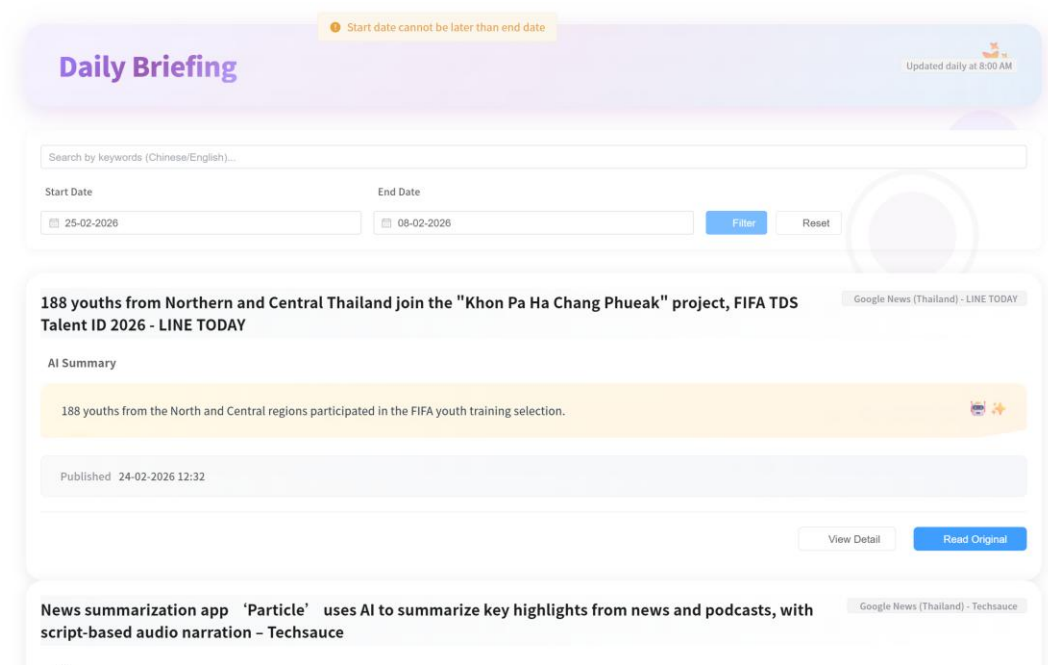
Document Name	BridgeU-SDD V.3.0	Owner	ZHIYI PAN	Page	82
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Chapter 4 UI Design

UI-01 Daily Briefing Page

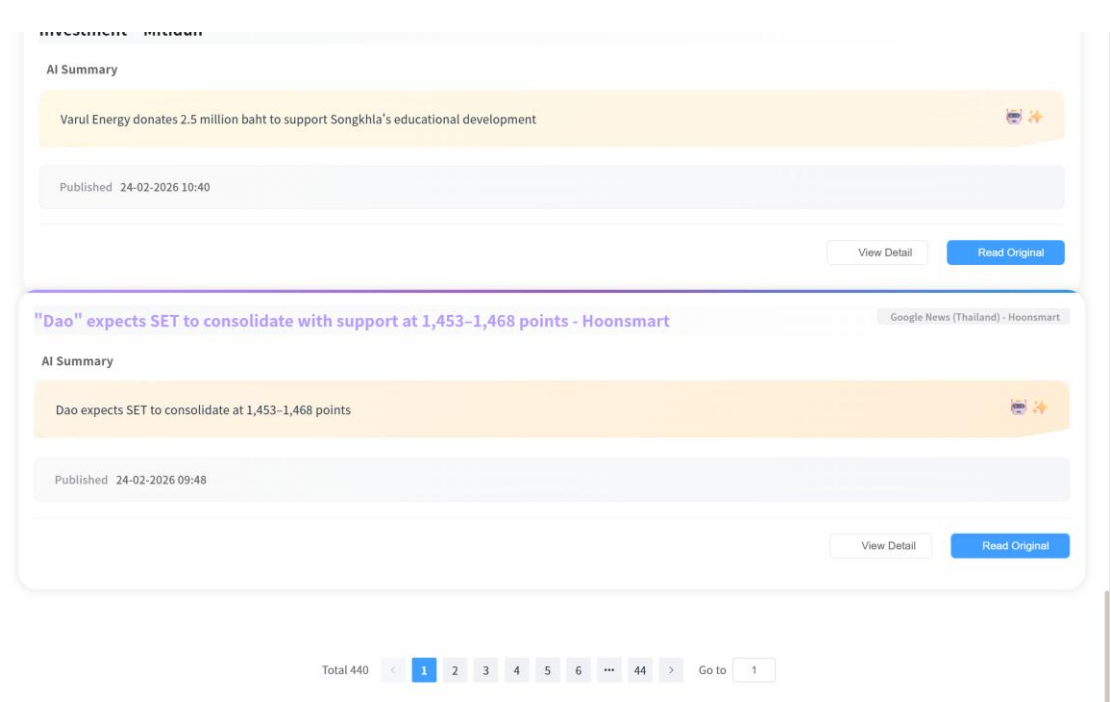


UI-01-01 Invalid date filtering

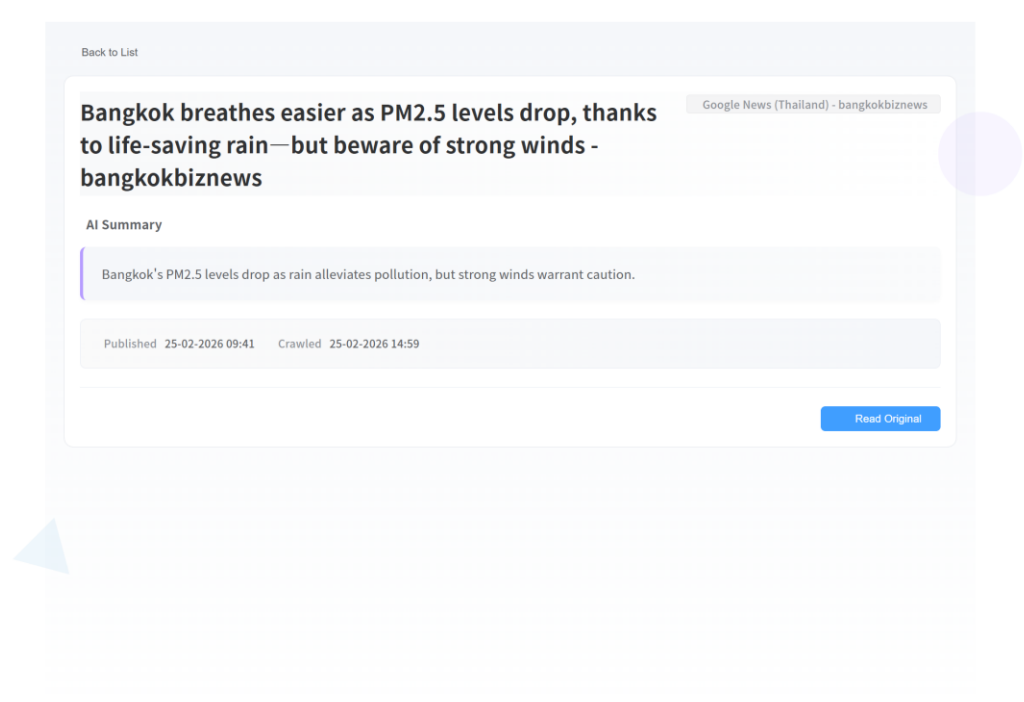


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UI-01-02 Select the page number

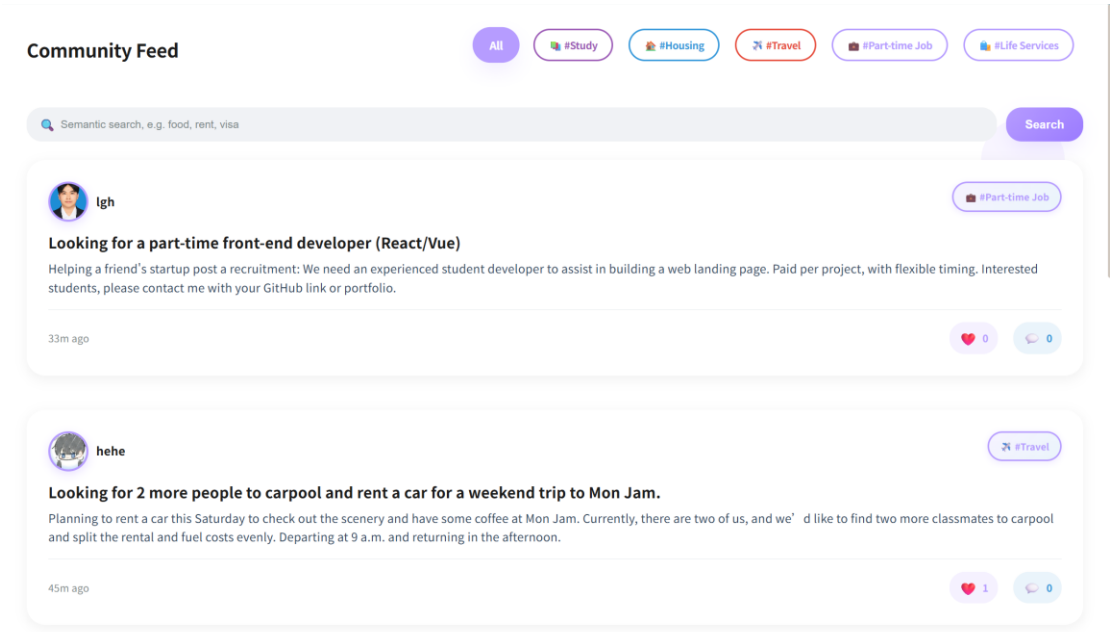


UI-02: Daily Briefing Page

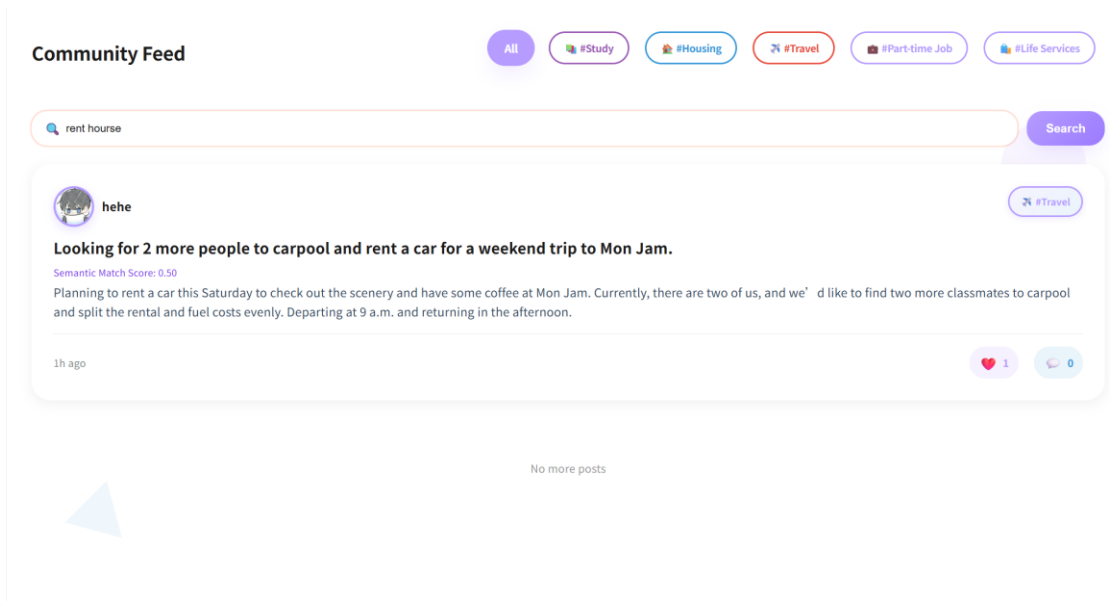


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UI-03:Community Page



UI-03-01:Search post



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UI-03-02:Filter post by tag

Community Feed

All

#Study

#Housing

#Travel

#Part-time Job

#Life Services

Semantic search, e.g. food, rent, visa

Search

pzyll

#Study

Seeking CMU Master of Software Engineering (MSE) interview experience

Just received the interview notification for this year’s Master’s in Software Engineering at Chiang Mai University. I’d like to ask seniors who’ve gone through the program before: what kinds of technical questions do professors usually ask? Do I need to specifically prepare an English self-introduction and details about my undergraduate thesis project?

1h ago

0

0

pzyll

#Study

Looking for teammates for the Kaggle "Spaceship Titanic" competition!

I'm currently working on the Spaceship Titanic dataset on Kaggle and am looking for 1-2 teammates to compete together. Ideally, you're familiar with Python and basic machine learning models (e.g., Random Forest, XGBoost). If you're interested, feel free to reach out—let's learn and collaborate together!

1h ago

0

0

UI-04:Post detail page

← Back

lgh

@lgh

Looking for a part-time front-end developer (React/Vue)

Helping a friend's startup post a recruitment: We need an experienced student developer to assist in building a web landing page. Paid per project, with flexible timing. Interested students, please contact me with your GitHub link or portfolio.

1

1

1h ago

Report

Add a comment

Write your comment...

Submit

Comments (1)

Summarize Comments

lgh

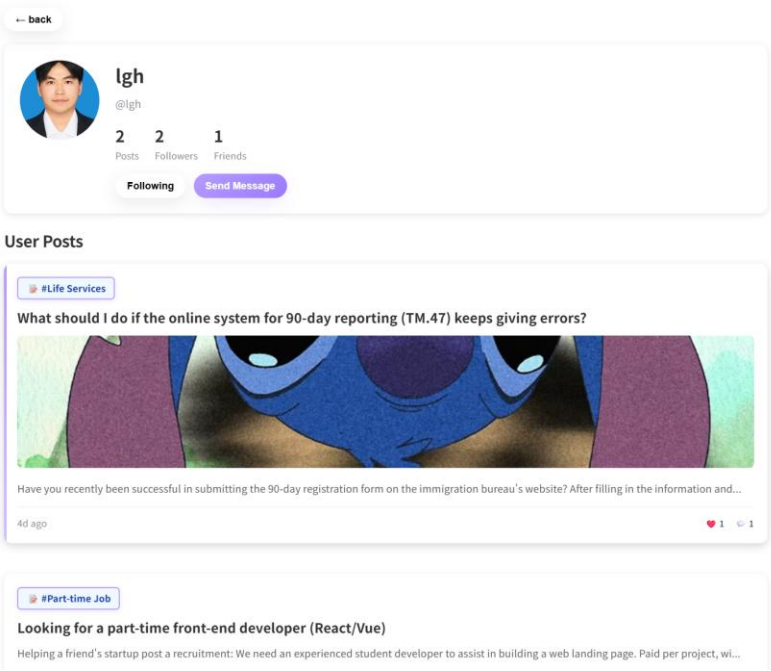
Just now

Everyone, sign up enthusiastically!

Delete

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UI-04—01:Other user’s profile page

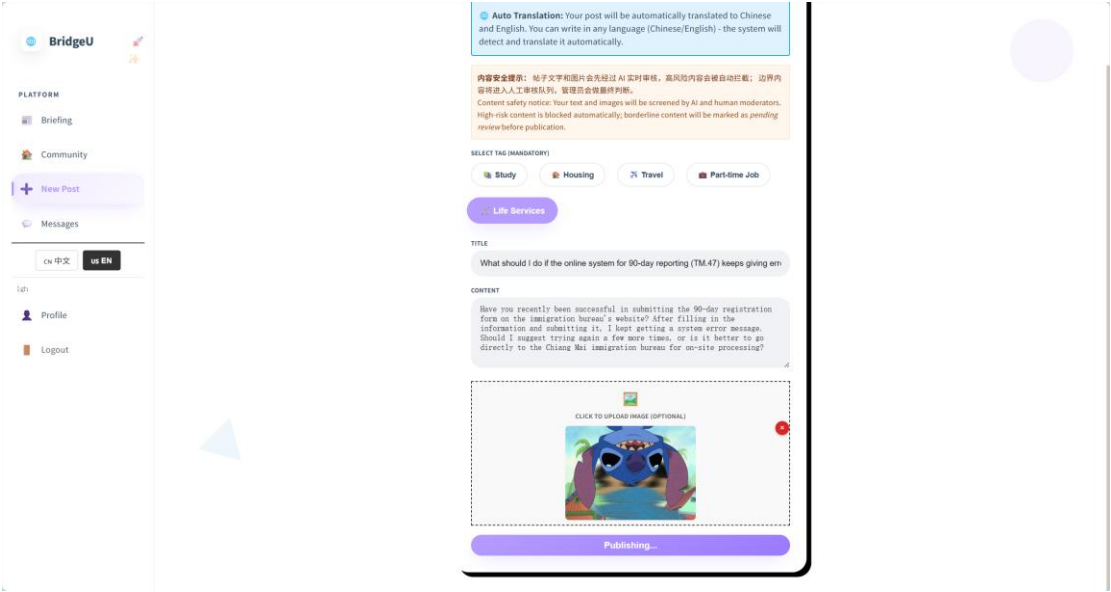


UI-05: Create Pubile Post Page

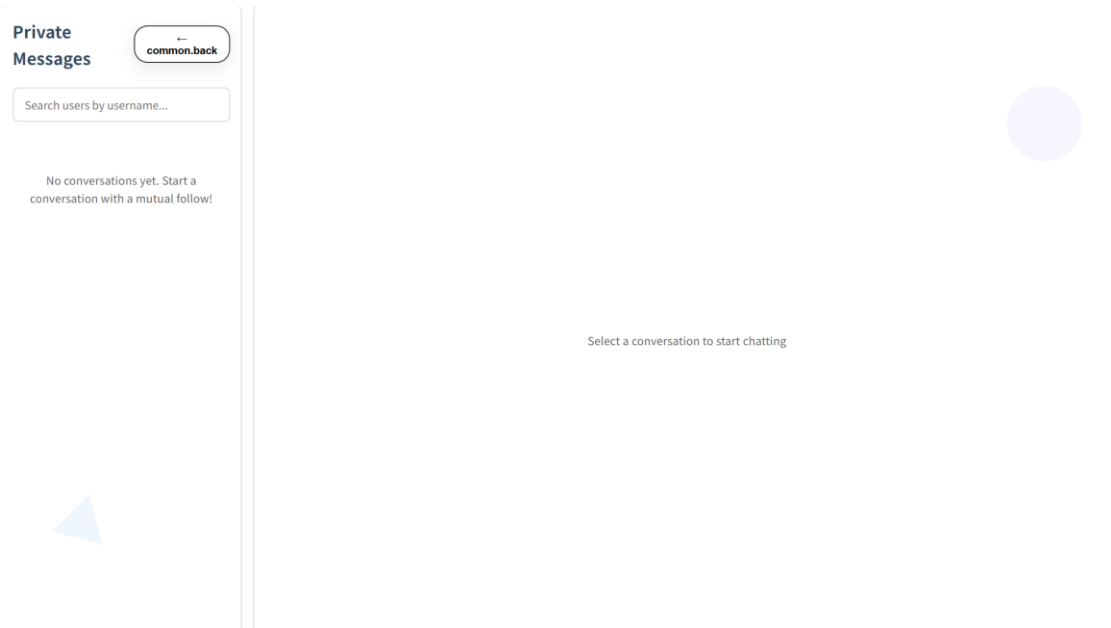


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UI-05-01:Creat Pubilc Post

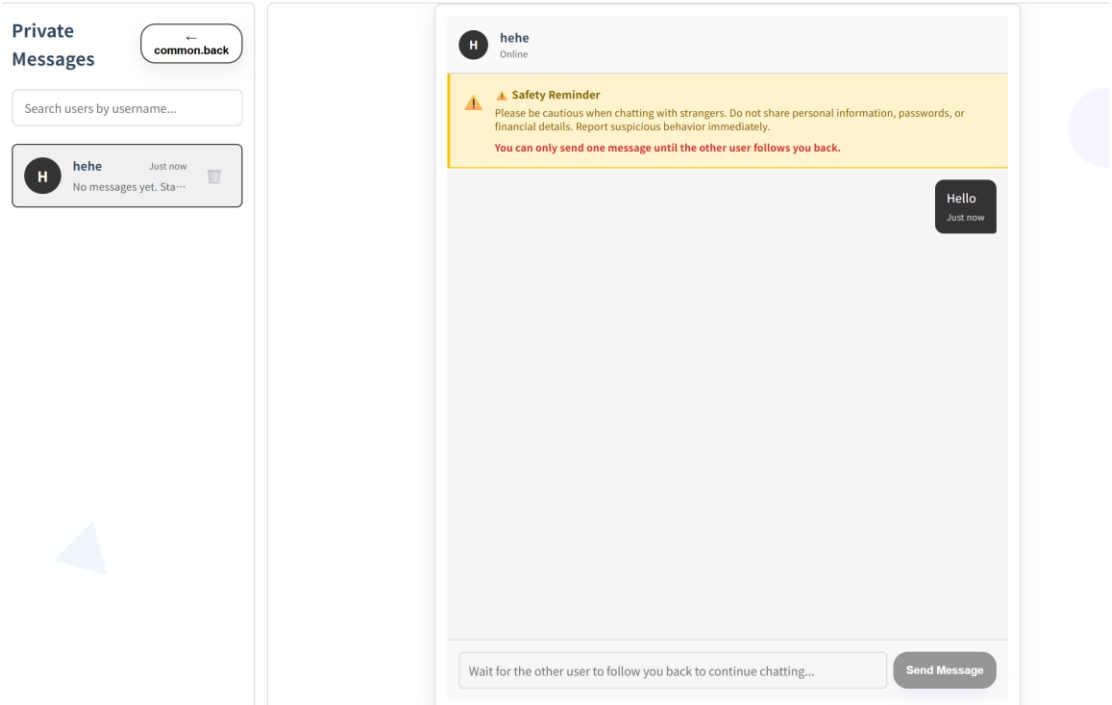


UI-06: Private Message Page

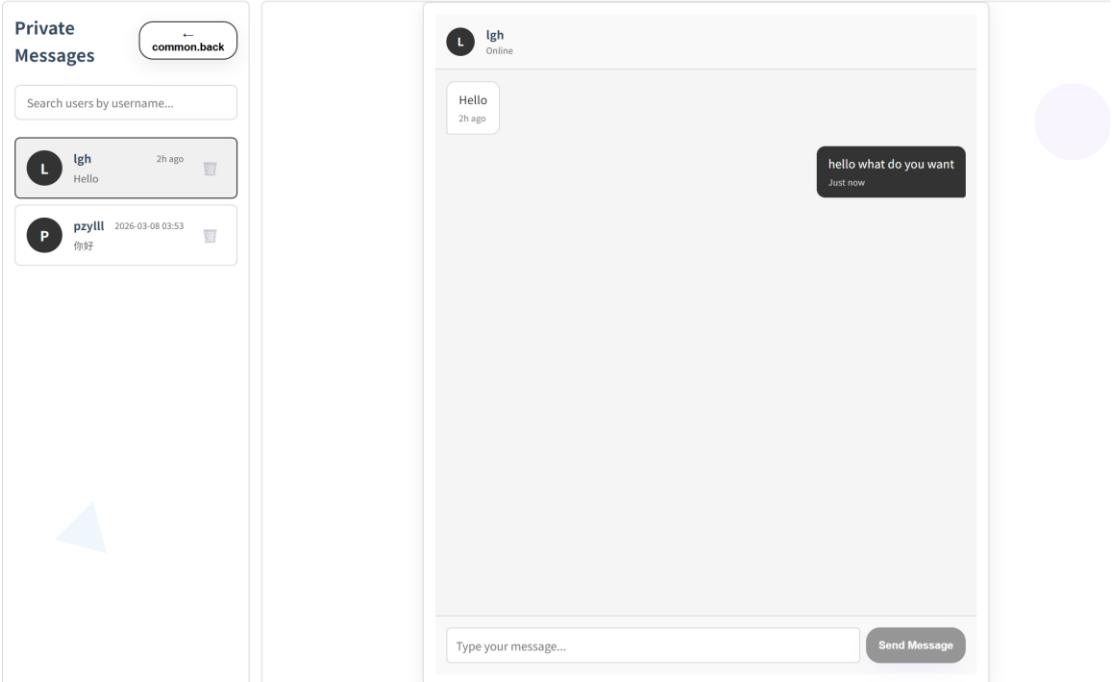


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UI-06-01: One-way following

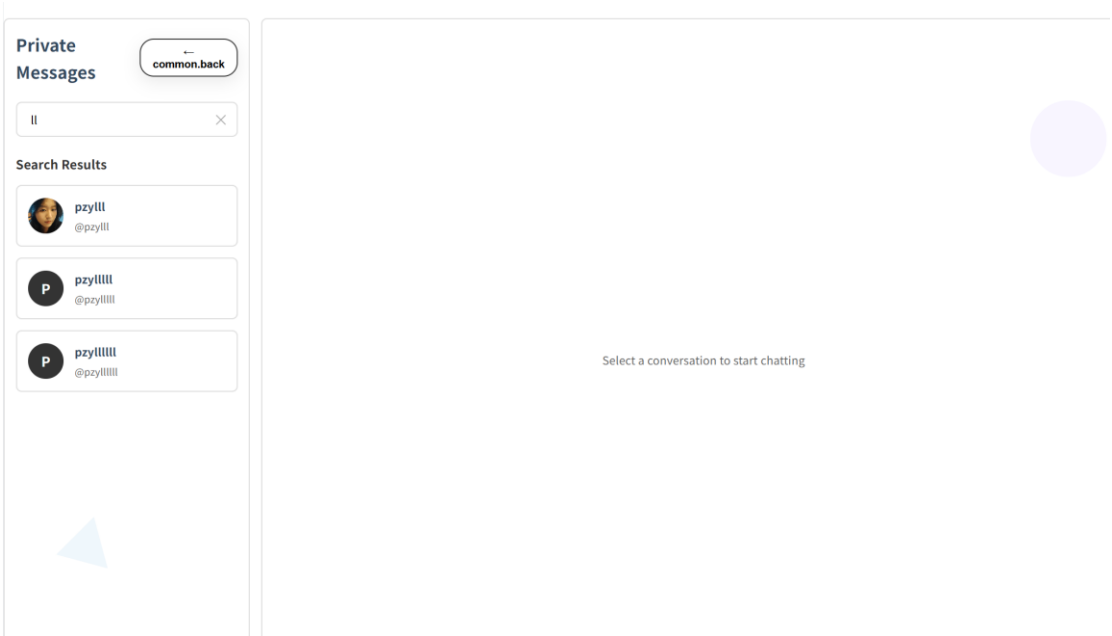


UI-06-02: Mutual Following

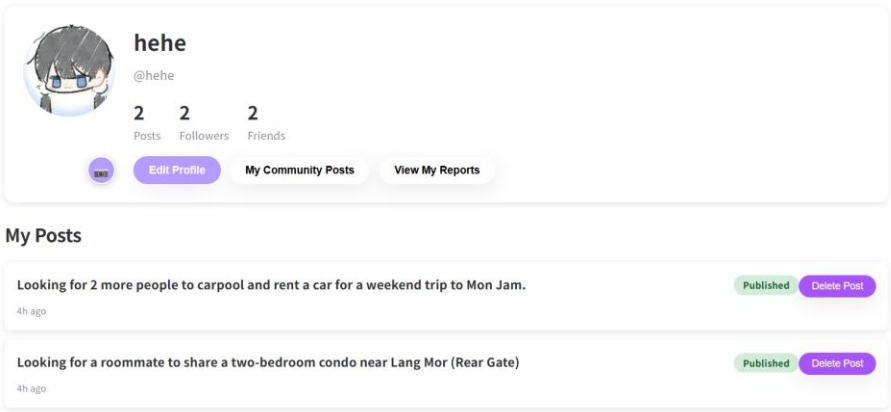


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UI-06-03:Search users by username

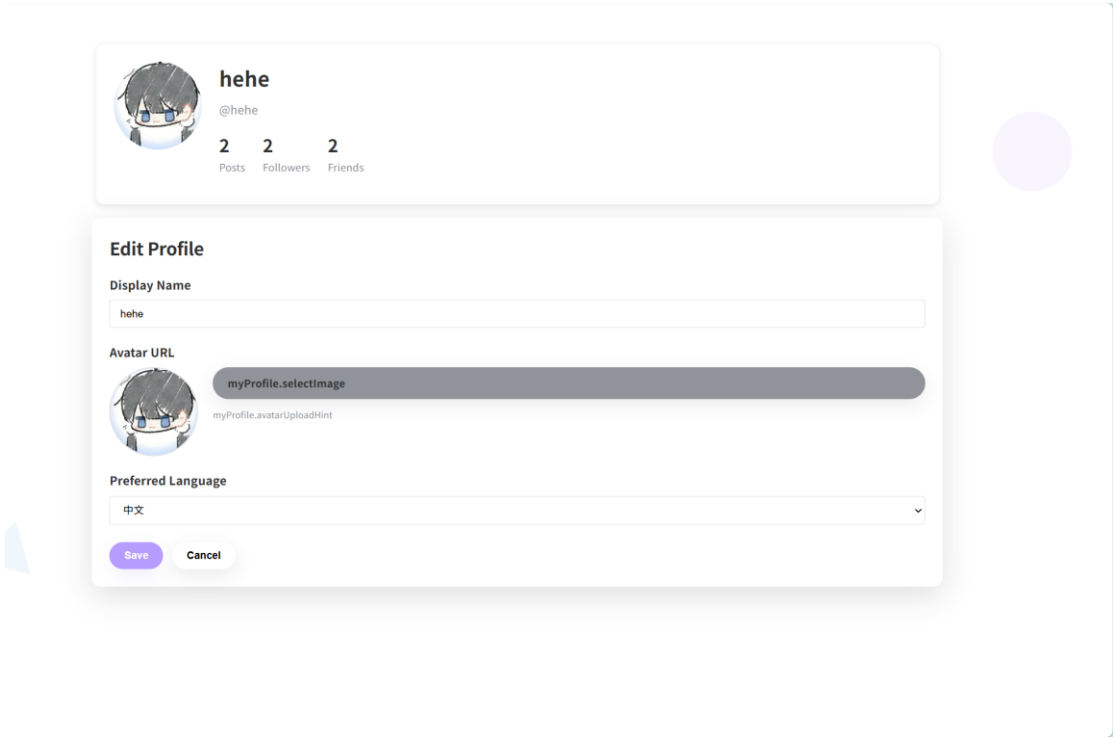


UI-07:Profile page

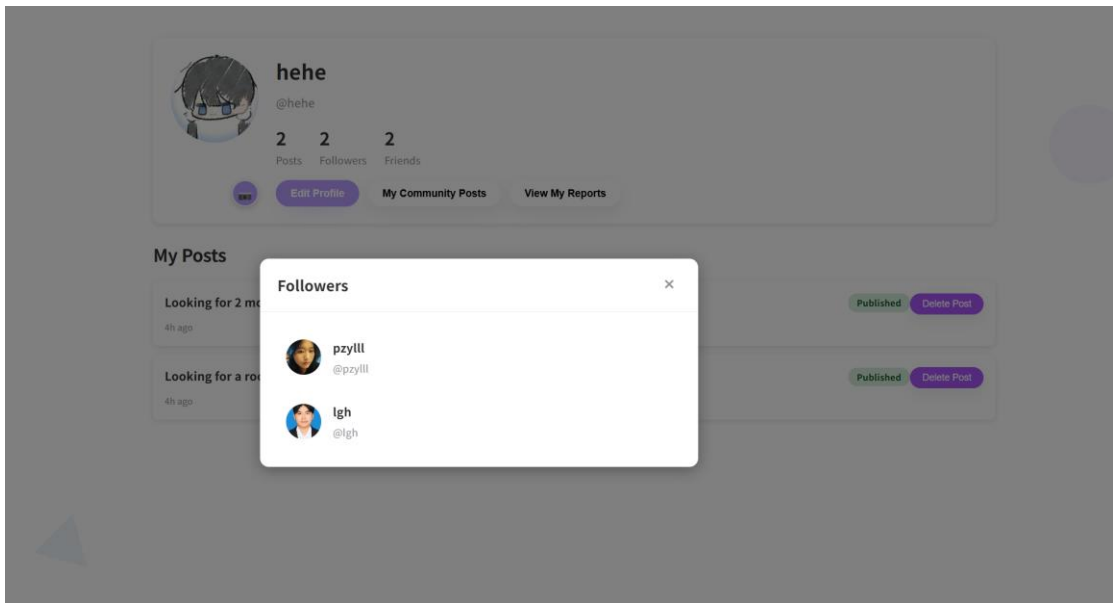


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UI-07-01:Edit Profile page

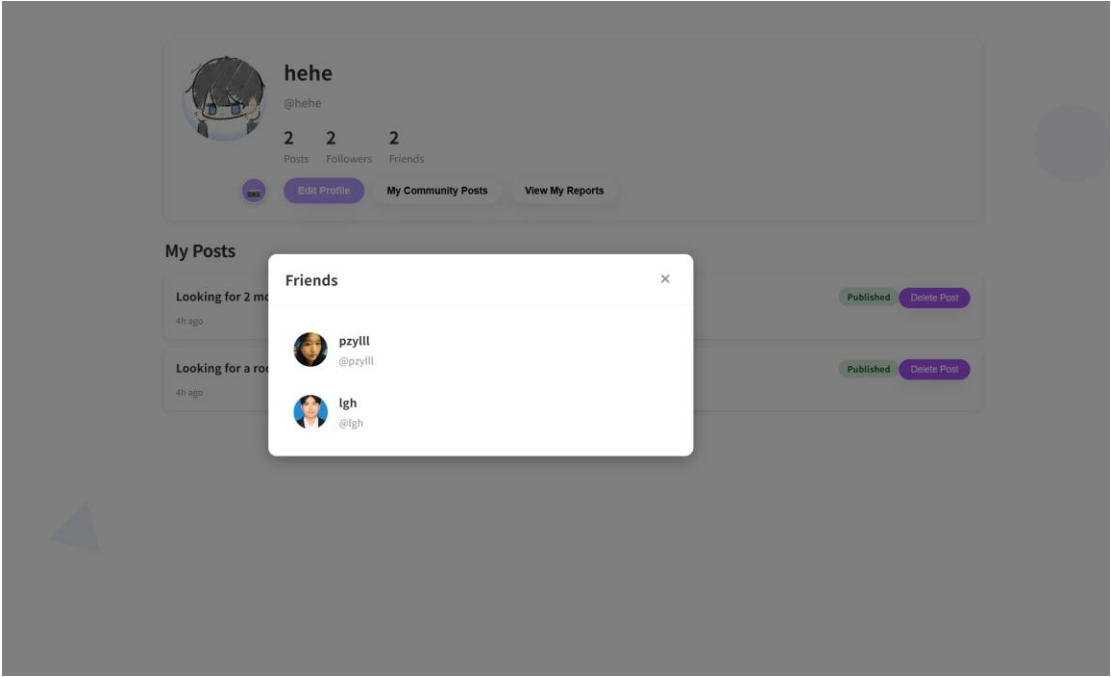


UI-07-02:Followers List Page

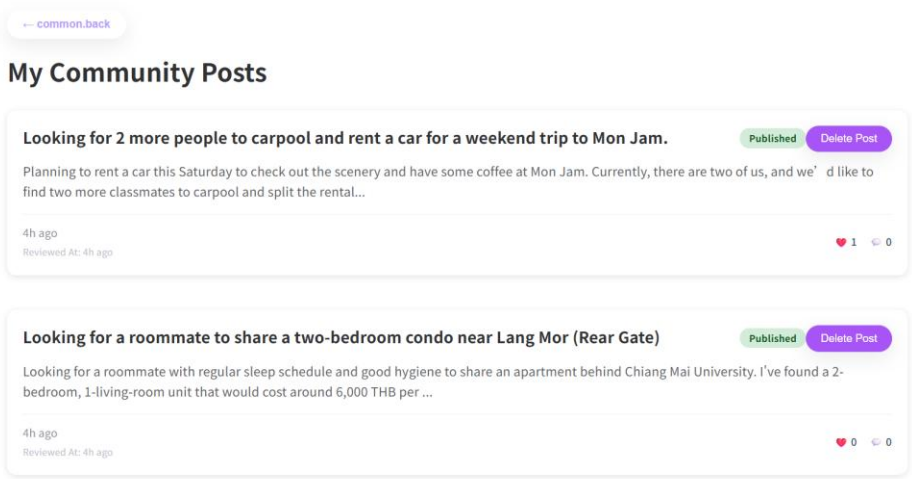


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UI-07-03:Fridens List Page



UI-08:My Community Posts Page



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UI-09:My Reports Page

← common.back

My Reports

Post

Dismissed

2026-03-14 22:17

Target ID:

42871f5d-492e-4477-a22b-c362f7de8e03

Reasons:

Spam

Review Notes

Review Date: 2026-03-14 22:17

No Violation

AI Confidence: 9500.0%

UI-10:Login Page

中文

BridgeU

Connecting Students & Local Life

USERNAME / EMAIL / PHONE NUMBER

USERNAME / EMAIL / PHONE NUMBER

PASSWORD

PASSWORD

LOGIN

Create an account

Forgot Password?

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UI-11:Forgot Password Page

BridgeU

Connecting Students & Local Life

Reset Password

Email

Enter your email address

Send Verification Code

New Password

New Password

Password must contain both uppercase and lowercase letters and be at least 6 characters

Confirm New Password

Confirm New Password

Reset Password

Back to Login

UI-12: Registration page

BridgeU

Connecting Students & Local Life

Complete Registration

Email

Enter your email address

Send Verification Code

Username

Username

Username must be English characters only

Password

Password

Password must contain both uppercase and lowercase letters and be at least 6 characters

Confirm Password

Confirm Password

Preferred Language

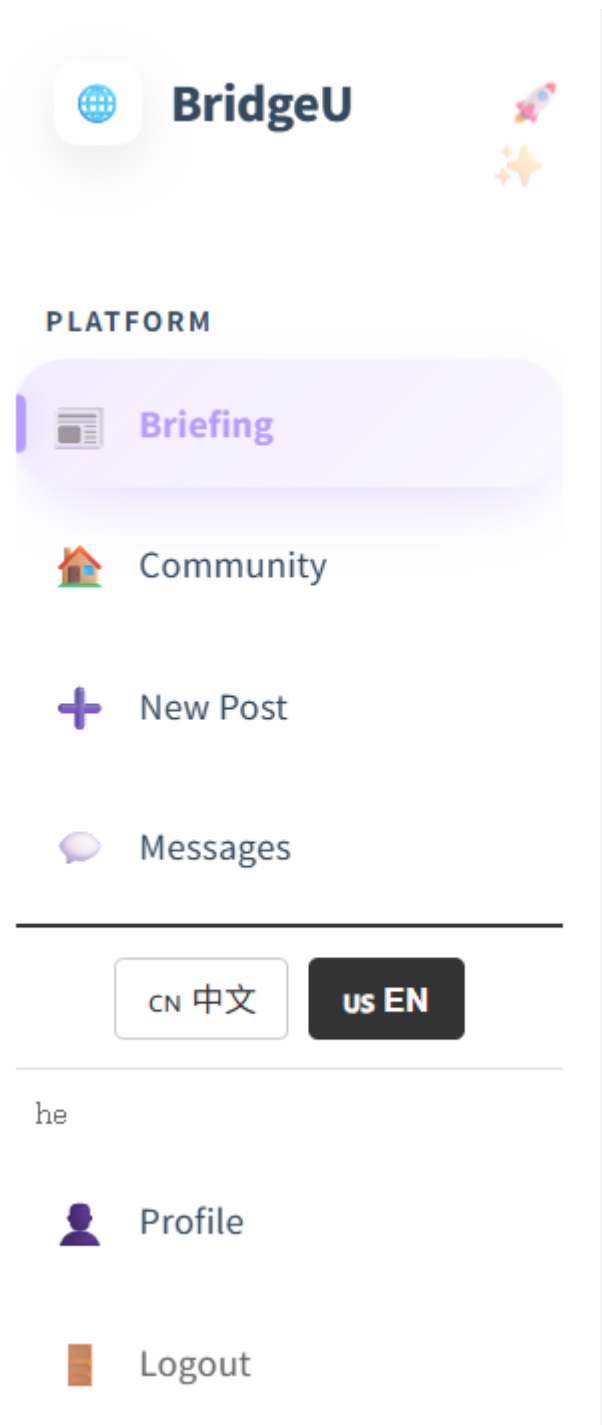
English

Register

Back to Login

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UI-13: Sidebar UI



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Chapter 5 Database Design

5.1 Entity Relationship Diagram (ERD)

The following diagram gives a **system-wide view** of all key entities in BridgeU and how they relate to each other. It mirrors the physical database schema used by the Spring Boot backend and can be read as follows:

- **USERS** :Core identity table for all roles (students, merchants, admins). Stores login credentials (`username`, `password\_hash`),contact info (`email`, `phone`), profile preferences (`preferred\_language`) and status flags (`enabled`, `created\_at`).
- **VERIFICATION_CODE** : Stores email/phone-based OTP codes for registration and password reset. `identifier` links logically to a user's email or phone, and each record includes the 6-digit `code`, type (`EMAIL`/`SMS`), purpose (`REGISTER`/`RESET\_PASSWORD`), expiration time (`expiresAt`) and usage flag (`used`).
- **POSTS / COMMENTS / POST_LIKES** – Represent community content and interactions. `POSTS` contains multilingual content and AI-moderation fields (`status`, `ai\_confidence`, `ai\_result`, `review\_note`); `COMMENTS` and `POST\_LIKES` reference `POSTS` and `USERS` to record who commented/liked what and when.
- **USER_FOLLOWS**: Many-to-many relation capturing who follows whom; used both for community features (mutual follow list) and as a permission gate for private messaging.
- **CONVERSATION / MESSAGE**: Model one-to-one private chats. `CONVERSATION` stores the pair of users and latest activity time,while `MESSAGE` stores individual messages, including sender/receiver, content, read flag and timestamps.
- **REPORT / REPORT_REASONS**`REPORT` stores user reports against posts or comments (`targetType`, `targetId`), including the selected reasons, free-text description, AI moderation output (`ai\_result`, `ai\_confidence`, `violation\_snippet`, `is\_violation`), status and review timestamps; `REPORT\_REASONS` stores the normalized list of selected reasons per report.
- **NOTIFICATIONS** – Stores user-facing notifications triggered by report processing and other events (e.g. report success/failure, Content penalties, restorations),

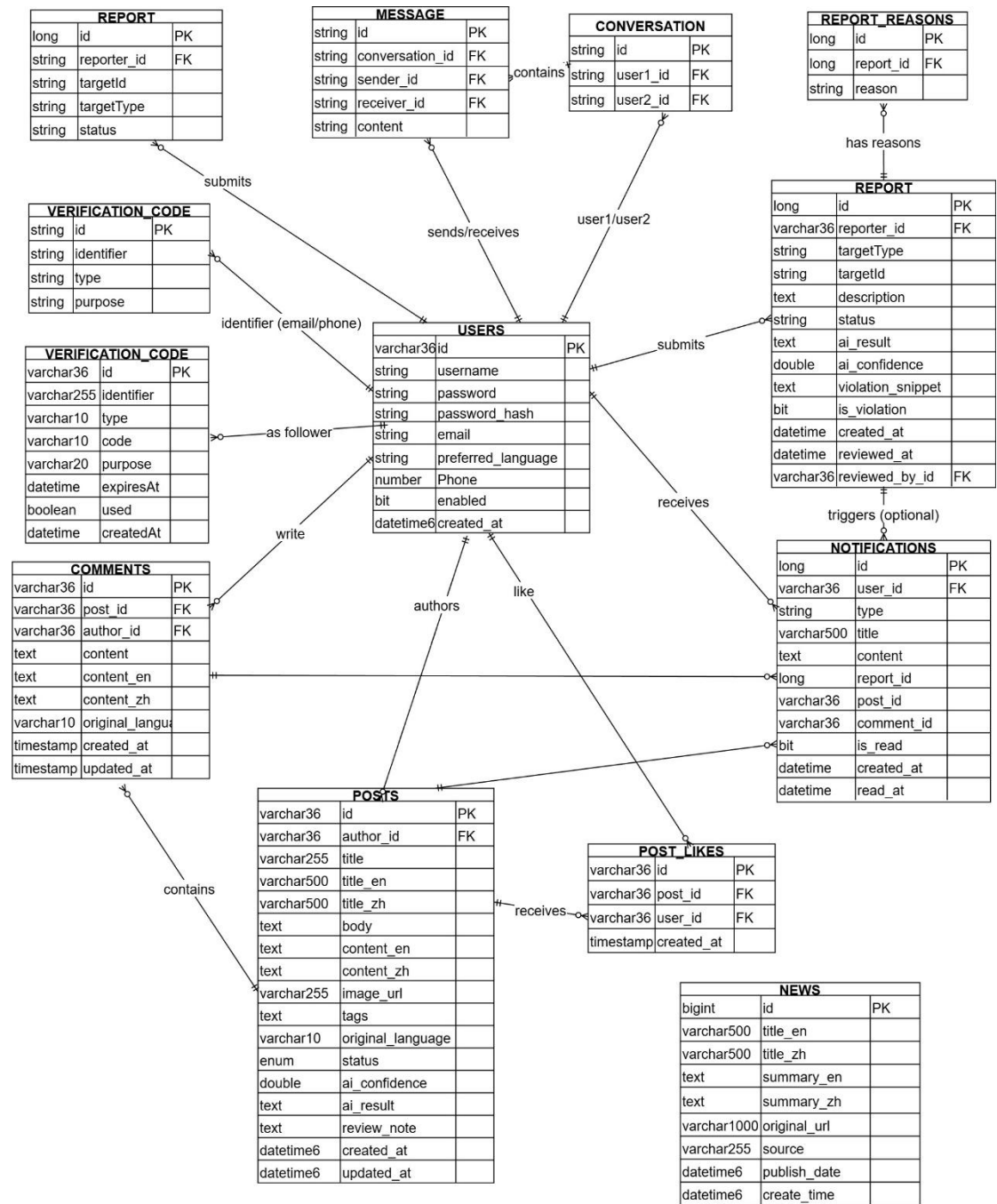
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including bilingual `title`/`content`, related `report\_id`/`post\_id`/`comment\_id`, and read/unread timestamps.

- **REPORT:** Stores user reports against posts or comments (`targetId`, `targetType`), together with status and timestamps, supporting the hybrid AI + human moderation flow.
- **NEWS:** Stores crawled Thai news and their bilingual titles/summaries, as used by the Daily Briefing feature(`title\_en/zh`, `summary\_en/zh`, `original\_url`, `source`, `publish\_date`, `create\_time`).

Together these entities support authentication, community interactions, messaging, content safety and the daily news briefing.

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5.2 Data Dictionary

Table Name	Field Name	Data Type	Description
Users	id	varchar(36) (PK)	Unique identifier for each user
	username	string	Login username
	password	string	plaintext password field if present
	password_hash	string	Hashed password used for authentication
	email	string	Email address
	preferred_language	string	UI language preference (zh/en)
	phone	number	Phone number
	enabled	bit	Whether the account is enabled
	created_at	datetime	Account creation timestamp
Verification_code	id	varchar(36) (PK)	Verification code record ID
	identifier	varchar(255)	Email address or phone number
	type	varchar(10)	Code type (EMAIL/SMS)
	purpose	varchar(20)	Code purpose (REGISTER/RESET_PASSWORD)
	code	varchar(10)	Verification code (6 digits)
	expiresAt	datetime	Expiration time of the verification code

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Table Name	Field Name	Data Type	Description
	used	boolean	Whether the code has already been used
	createdAt	datetime	Creation time of the verification code record
Posts	id	varchar(36) (PK)	Post ID
	author_id	varchar(36) (FK)	References `USERS.id` (post author)
	title	varchar(255)	Original title
	title_en	varchar(500)	English title
	title_zh	varchar(500)	Chinese title
	image_url	varchar(255)	Optional image URL
	tags	text	Tag list
	original_language	varchar(10)	Detected original language(zh/en)
	status	enum	Moderation status (PENDING_REVIEW/APPROVED/REJECTED)
	ai_confidence	double	AI moderation confidence score
	ai_result	text	AI moderation result payload
	review_note	text	AI review note
	created_at	datetime	Creation time

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Table Name	Field Name	Data Type	Description
	updated_at	datetime	Update time
Comments	id	varchar(36) (PK)	Comment ID
	post_id	varchar(36) (FK)	References `POSTS.id`
	author_id	varchar(36) (FK)	References `USERS.id`
	content	text	Original comment content
	content_en	text	English comment content
	content_zh	text	Chinese comment content
	original_language	varchar(10)	Detected original language
	created_at	timestamp	Creation time
	updated_at	timestamp	Update time
Post_Likes	id	varchar(36) (PK)	Like record ID
	post_id	varchar(36) (FK)	References `POSTS.id`
	user_id	varchar(36) (FK)	References `USERS.id`
	created_at	timestamp	Like time
User_Follows	id	varchar(36) (PK)	Follow relationship ID

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Table Name	Field Name	Data Type	Description
	follower_id	varchar(36) (FK)	References `USERS.id` (follower)
	following_id	varchar(36) (FK)	References `USERS.id` (followed user)
	created_at	timestamp	Follow time
Conversat ion	id	string(PK)	Conversation ID
	user1_id	string(FK)	References `USERS.id` (participant 1)
	user2_id	string(FK)	References `USERS.id` (participant 2)
	created_at	datetime	Conversation creation time
	last_message_at	datetime	Timestamp of the latest message
Message	id	string(PK)	Message ID
	conversation_id	string(FK)	References `CONVERSATION.id`
	sender_id	string(FK)	References `USERS.id`
	receiver_id	string(FK)	References `USERS.id`
	content	text	Message content
	is_read	bit	Read flag
	read_at	datetime	Read time
	created_at	datetime	Created time

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Table Name	Field Name	Data Type	Description
News	id	bigint (PK)	News ID
	title_en	varchar(500)	English title
	title_zh	varchar(500)	Chinese title
	summary_en	text	English summary
	summary_zh	text	Chinese summary
	original_url	Varchar (1000)	Original URL
	source	varchar(255)	Source website name
	publish_date	datetime	Publication time
	create_time	datetime	Insert time (when stored)
Reports	id	Long(PK)	Report ID
	reporter_id	varchar(36) (FK)	References USERS.id (user who submitted the report)
	targetType	string	Target type: POST or COMMENT
	targetId	string	Target entity ID (post ID or Comment ID)
	status	string	Report status (PENDING, RESOLVED, DISMISSED)
	description	text	Optional free-text description from the reporter

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Table Name	Field Name	Data Type	Description
	ai_result	text	Raw JSON string returned by the AI moderation model
	ai_confidence	double	AI confidence score in the range [0.0, 1.0]
	violation_snippet	text	Concrete snippet that violates the guidelines (as extracted by AI)
	is_violation	bit	Whether the reported content is considered a violation (true/false)
	created_at	datetime	Timestamp when the report was created
	reviewed_at	datetime	Timestamp when the report was reviewed (AI or admin)
	reviewed_by_id	varchar(36) (FK)	References USERS.id (admin who performed the final review, if any)
Report_reasons	id	Long(PK)	Internal identifier of a single report reason entry
	report_id	Long(FK)	References REPORTS.id (owning report)
	reason	string	One of the predefined reasons (Spam, Fraud or Scam, Abusive Language etc.)
Notifications	id	Long(PK)	Notification ID
	user_id	varchar(36)(FK)	References USERS.id (receiver of the notification)

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Table Name	Field Name	Data Type	Description
	type	string	Type (REPORT_SUCCESS, REPORT_FAILED, POST_VIOLATION_PENALTY, POST_RESTORED, etc.)
	title	varchar(500)	Notification title (often stored as bilingual text)
	content	text	Notification body content (often stored as bilingual text)
	report_id	long	Related report ID when applicable (logical link to REPORTS.id)
	post_id	varchar(36)	Related post ID when applicable (logical link to POSTS.id)
	comment_id	varchar(36)	Related comment ID when applicable (logical link to COMMENTS.id)
	is_read	bit	Whether the notification has been read by the user
	created_at	datetime	Creation time of the notification
	read_at	datetime	Timestamp when the notification was marked as read

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